

Inflation expectations and inflation realities: a comparison of the Treasury Breakeven Inflation curve and the Consumer Price Index before, during, and after the Great Recession

This article examines the extent to which market-based inflation expectations overshoot or undershoot actual inflation before, during, and after the Great Recession of 2007–09. Specifically, the article compares the U.S. Treasury Breakeven Inflation (TBI) curve, a unique measure of market-based inflation expectations, with the U.S. city-average, all-items Consumer Price Index for All Urban Consumers (CPI-U), a measure of actual inflation. Covering the 175 months from July 2003 through January 2018, the analysis yields three main findings. First, market-based inflation expectations, as measured by the TBI curve, reasonably approximated CPI-U (realized) inflation in the years before, during, and after the recession. Second, estimates of inflation expectations overshoot actual inflation for short-term maturity horizons and undershoot it for long-term maturity horizons. Third, inflation expectations approximated inflation reality more precisely for long-term rates than for short-term rates.



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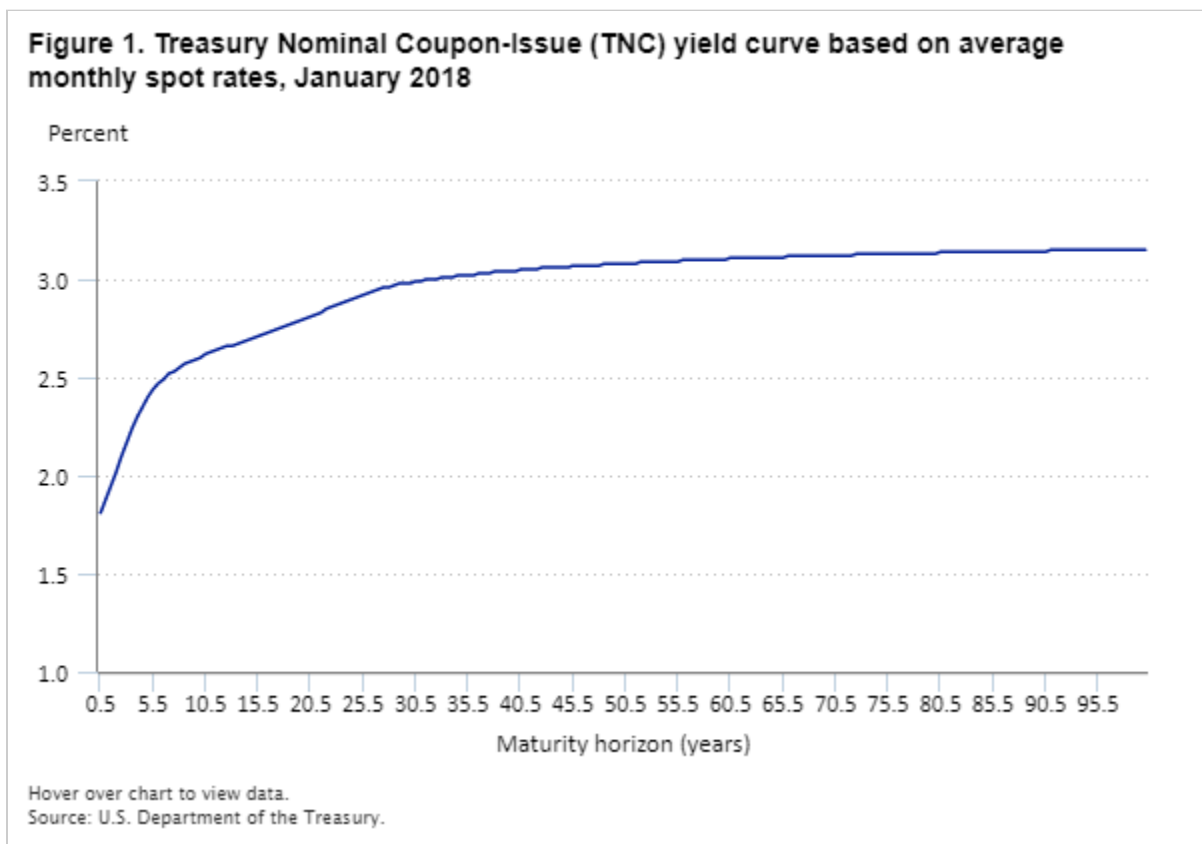
On January 29, 1997, the U.S. Treasury Department issued a 10-year inflation-indexed Treasury note and announced that, for the foreseeable future, 10-year Treasury Inflation-Protected Securities (TIPS) would be regularly auctioned on the 15th of January, April, July, and October of each year.¹ In subsequent years, 5-, 20-, and 30-year notes were also introduced. Currently, the U.S. Treasury Department issues 5-, 10-, and 30-year notes, all indexed to the Consumer Price Index for All Urban Consumers (CPI-U).²

In 1996, a year before the U.S. Treasury Department held its first auction of 10-year notes, John Y. Campbell and Robert J. Shiller discussed the primary issues motivating interest in inflation-indexed debt.³ Among these were (1) the effect of inflation-indexed debt issuance on government borrowing costs, (2) the relative risk-sharing efficiency

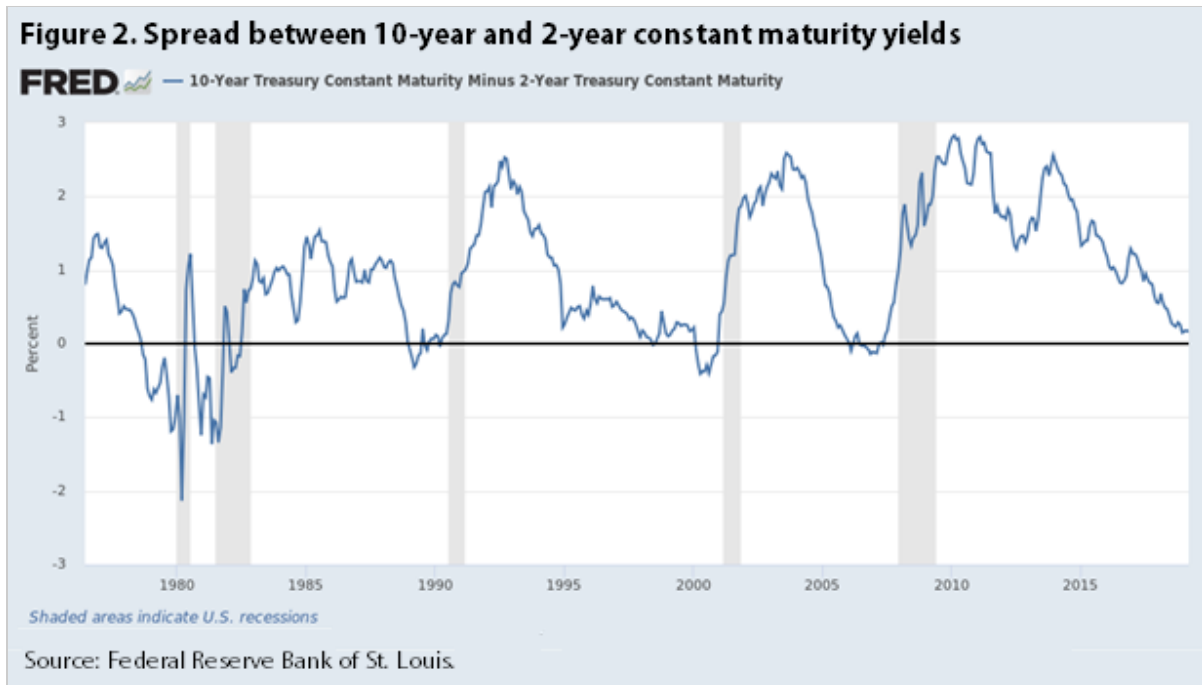
of inflation-indexed debt as an inflation-hedging asset, and (3) the use of inflation-indexed debt to derive market-based inflation expectations.

This article addresses the third issue. The market-based measure of inflation expectations is widely known as the breakeven rate of inflation. There are several methods to calculate the yield curve and, thus, capture the breakeven rate. As explained in more detail below, this article uses the Treasury Breakeven Inflation (TBI) curve, which is based on the nominal and real yield curves calculated by the U.S. Treasury Department. Although TBI rates are a secondary benefit of inflation-indexed debt (TIPS are primarily seen as an inflation hedge and a potentially cheaper source of debt financing), they are nonetheless viewed as a source of information on inflation expectations derived from the market activities of investors.

The breakeven rate of inflation is derived from the Treasury yield curve. The yield curve, a fundamental concept in finance, refers to the spread between long- and short-term yields on Treasury securities (e.g., the spread between 10- and 2-year Treasury notes), ultimately reflecting various expectations about long- and short-term risks to the economy and financial markets. In an economy with a reasonably positive outlook, the yield curve slopes upward, indicating greater uncertainty about longer term rates. This typical case is shown in figure 1, which graphs average monthly Treasury spot rates (derived from yields, a distinction explained in more detail below) calculated for maturities in increments of 6 months, up to 100 years, as of January 2018.



Instead of directly graphing yields (or, in the above case, spot rates), one can represent the yield curve as a graph showing the spread between, for example, the 10- and 2-year Treasury constant maturity yields. This representation is illustrated in figure 2.



The yield curve also can slope downward, and when it does, it often signals the onset of a recession. Indeed, in her historical survey of the yield curve, Caitlin Zaloom states that “[a]n ‘inverted’ curve has preceded each recession since the mid-1960s (with one exception), a record that some use to orient their strategies, while others question its salience.”⁴ A downward-sloping yield curve indicates that investors expect long-term rates to decline as the economy enters into a recession, the demand for loans declines, and the U.S. Federal Reserve Board reduces rates to stimulate the economy.

Besides predicting recessions, the yield curve can predict inflation, providing a benchmark for investor expectations about future inflation. This relationship is based on the so-called Fisher effect (named after economist Irving Fisher), which is commonly represented as an equation that sets the real interest rate equal to the nominal interest rate minus inflation.⁵ The intuition behind the equation is based on the observation that, to arrive at the real return on a dollar of investment, one must discount the nominal return on that dollar by the rate of inflation. Because inflation measures the change in purchasing power on a dollar of nominal return, the breakeven inflation rate is derived as a discount rate that equates the nominal return on a dollar of investment to the real return (not as an arithmetic difference between the nominal and real rates). For example, one can use the nominal and real yield curves to obtain, respectively, the nominal and real yields on 1-, 2-, and 10-year Treasury securities. From these yields, one can derive the 1-, 2-, and 10-year nominal and real spot rates. These spot rates can then be used to derive forward rates, which reflect rates that would prevail over a specified horizon beyond some future point.

Once yields, spot rates, or forward rates are obtained, breakeven inflation rates can be derived as annualized rates from the nominal and real rates of return (yields or spot rates) on Treasury securities. Consistent with market convention, nominal and real rates of return on fixed-income securities are calculated semiannually, while inflation rates are calculated annually. Thus, the TBI curve converts semiannually compounded nominal and real rates to annual rates, which can then be compared with annual CPI-U inflation rates. The TBI rate is the expected annualized rate of inflation and is calculated as

$$\left[\frac{(1 + \text{nominal spot rate}/2)}{(1 + \text{real spot rate}/2)} \right]^2 - 1 \times 100,$$

where the nominal and real spot rates are the monthly average nominal and real spot rates published by the U.S. Treasury Department.⁶

Breakeven rates may have an upward bias because of a term premium (embedded in the nominal yield) that reflects uncertainty about future inflation. Alternatively, breakeven rates may have a downward bias because of a term premium that reflects the relative illiquidity of TIPS. Moreover, in practice, nominal and real yields tend to be lower than their respective spot rates, and this difference creates a downward bias in breakeven rates when one uses yields instead of spot rates.⁷ Finally, market volatility implies that expectations will not always match reality. Nonetheless, investors view TBI rates as a barometer of future inflation rates.

Literature review

A vast literature studies the dynamics of inflation. Since the emergence of inflation-indexed sovereign debt in the United States, a segment of this literature has examined an observed discrepancy between, on the one hand, the breakeven rate of inflation and survey-based measures of inflation expectations, and, on the other, the breakeven rate of inflation and the realized rate of inflation. Not long after the initial issuance of inflation-indexed debt by the U.S. Treasury Department, economists began to observe that TIPS breakeven rates frequently underestimated both survey-based forecasts of inflation and realized rates of consumer price inflation.

The literature focuses primarily on two risk premiums embedded in Treasury securities: (1) the inflation risk premium embedded in the nominal yield on ordinary Treasury securities, and (2) the liquidity risk premium embedded in the real yield on TIPS. In a 2013 journal article, Olesya V. Grishchenko and Jing-Zhi Huang note the lack of consensus on the size of the inflation risk premium and on whether that premium is positive or negative, which often depends on economic circumstances.⁸ However, in a 2010 study of inflation-hedging strategies based on various asset classes, Geert Bekaert and Xiaozheng Wang present historical evidence that the inflation risk premium tends to be positive, reflecting compensation to investors for bearing the risk of inflation volatility.⁹

A positive inflation risk premium increases the nominal yield on ordinary Treasury securities or, alternatively, decreases the real yield on TIPS (because investors effectively pay “insurance” to avoid inflation risk). All else equal, a lower TIPS yield (or, alternatively, a higher nominal yield on ordinary Treasury securities) leads the breakeven rate of inflation to overshoot realized inflation.

The liquidity risk premium also tends to be positive, reflecting compensation to investors for bearing the risk associated with TIPS having a lower level of liquidity than ordinary Treasury securities. TIPS are less liquid because they lack the market depth of ordinary Treasury securities, are typically held by buy-and-hold investors seeking an inflation hedge, and, in the early years of their issuance, bore the uncertainty of whether the U.S. Treasury Department would continue to issue inflation-indexed debt. A positive liquidity risk premium increases the real yield on TIPS, and, all else equal, this increase leads the breakeven rate of inflation to undershoot realized inflation.

If the inflation risk premium and the liquidity risk premium are equal, they cancel each other and the breakeven rate of inflation approximates realized inflation (assuming other market factors driving the supply and demand of government debt remain constant). However, because breakeven rates have been found to consistently

underestimate survey-based inflation expectations and actual inflation, the literature on TIPS has often focused on estimating the liquidity premium.¹⁰

It should be noted that breakeven rates are only one method for forecasting inflation. In a 2007 article, Andrew Ang, Geert Bekaert, and Min Wei discuss four alternative methods: “time-series ARIMA models; regressions using real activity measures motivated from the Phillips curve; term structure models that include linear, nonlinear, and arbitrage-free specifications; and survey-based measures.”¹¹ In general, the authors conclude that surveys generate the best forecasts, especially in comparison with term structure models, and that combining forecasts from different approaches does not improve the results.

The Treasury Breakeven Inflation curve

The economic literature typically focuses on several theoretical and empirical aspects of inflation-indexed debt, using a variety of modeling techniques to forecast inflation. By contrast, the present article is concerned with the efficacy of TIPS breakeven rates—specifically those calculated from the Treasury Breakeven Inflation (TBI) curve—as a measure of market-based inflation expectations. The analysis uses monthly breakeven rates, calculated for short- and long-term maturity horizons by the U.S. Treasury Department, to examine how well market-based inflation expectations matched realized inflation before, during, and after the Great Recession. To my knowledge, this is the first attempt to use the TBI curve to examine the efficacy of breakeven inflation rates on a monthly basis.

The TBI curve is derived from nominal and real rates of return on off-the-run, coupon-issue Treasury notes and bonds—or, more specifically, from the Treasury Nominal Coupon-Issue (TNC) and Treasury Real Coupon-Issue (TRC) yield curves.¹² The TNC curve is derived by adjusting for a hump in yields observed at the 20-year maturity horizon and for the price difference between on-the-run and off-the-run securities. The TRC curve is derived from returns on TIPS, on the basis of a methodology similar to that used for deriving the TNC curve. However, while the TRC methodology adjusts for the hump in yields at the 20-year maturity horizon, it does not distinguish between on-the-run and off-the-run securities.¹³ The TBI curve shows monthly breakeven inflation rates derived from the TNC and TRC curves, using incremental 6-month maturity horizons.

Breakeven rates can be obtained directly from the Treasury yield curves, from spot rates derived from the Treasury yield curves, or from forward rates derived from spot rates.¹⁴ Despite some debate about which measure is most appropriate, the use of spot rates is generally the best approach for arriving at a “prediction” of future inflation. Because spot rates equate a one-time future payment (e.g., a principal repayment) to its present value (i.e., the rate of return on a zero-coupon bond), they avoid the complexities associated with yields or, more specifically, the yield to maturity (YTM). The YTM is an internal rate of return that equalizes the cash outflow resulting from the purchase of a Treasury security (or any fixed income security) with the cash inflow generated by coupon payments (assumed to be reinvested at a fixed rate) and the repayment of principal at the end of term. It is important to note, however, that forward rates can also be used to predict future inflation.

In practice, in making inflation predictions, investors and the financial press often rely on breakeven inflation rates derived from both yields and spot rates. Either way, TBI rates, derived from the TNC and TRC curves (which use spot rates), are increasingly used as a measure of inflation expectations. For example, in computing lump-sum retirement payouts, the Uniformed Services Blended Retirement System of the U.S. Department of Defense uses the U.S. Treasury Department’s breakeven inflation rates.¹⁵

Matching TBI rates to their respective annualized CPI-U rates

To measure actual inflation, this analysis relies on annualized rates of change in the nonseasonally adjusted U.S. city-average, all-items Consumer Price Index for All Urban Consumers (CPI-U) (1982–84 = 100). The CPI-U measures the average change over time in the prices paid by urban consumers for a constant-quality market basket of goods and services (i.e., a sample of goods and services purchased by consumers). Produced monthly, the CPI-U weights the price of each item in the market basket by the amount of spending reported by a sample of families and individuals. It also reflects additional adjustments, such as imputation (to account for missing quotes) and hedonics (to account for changes in item quality).¹⁶

This analysis uses the CPI-U because the real yields on Treasury securities are derived from TIPS, which adjust principal and coupon payments by the rate of change in the nonseasonally adjusted CPI-U. Thus, the real yield is equal to the inflation-adjusted coupon payments divided by the market price of Treasury securities.

TBI rates are mapped to their respective annualized CPI-U inflation rates. For example, the 6-month TBI rate in January 2003 (0.39 percent) is matched to the annualized CPI-U inflation rate (2.44 percent) from January to July 2003 (i.e., the squared 6-month rate of change in the CPI-U between January 2003 and July 2003). Because the TBI data begin in January 2003, the first month in which a TBI rate is available for matching is July 2003 (for the rate derived from 6-month spot rates on Treasury securities in January 2003), the second month is August 2003 (for the rate derived from 6-month spot rates on Treasury securities in February 2003), and so on, until December 2003. In January 2004, two TBI rates become available for matching (the rate derived from 6-month spot rates on Treasury securities in July 2003 and the rate derived from 1-year spot rates on Treasury securities in January 2003), and the same holds for all months until June 2004. In July 2004, three TBI rates become available for matching.

Given that the TBI data begin in January 2003, CPI-U rates published from July to December 2003 can only be compared with TBI rates derived from spot rates on Treasury securities with a 6-month maturity from January to June 2003. TBI rates are unavailable for matching when they are derived from spot rates on Treasury securities with a 1-year maturity from July to December 2002, from spot rates with a 1.5-year maturity from January to June 2002, and so on.

Because TBI data are unavailable before January 2003, more TBI rates are available for matching in later than earlier years. For example, there are 10 years of TBI rates available for matching in January 2013, which, given that TBI rates are available in 6-month increments, equates to 20 TBI rates. These rates are based, respectively, on the 6-month spot rates on Treasury securities in July 2012, on the 1-year spot rates on Treasury securities in January 2012, and so on, back to January 2003.

Given that TBI rates for longer maturity horizons are available only for their annualized CPI-U inflation rates in later months, the number of observations of TBI rates derived from spot rates on Treasury securities with a shorter maturity (e.g., 6 months) is greater than the number of TBI rates derived from spot rates on Treasury securities with a longer maturity (e.g., 1 year). (See table 1.)

Table 1. Difference between expected inflation, as measured by TBI rates, and actual inflation, as measured by CPI-U inflation rates, for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018

Maturity horizon (years)	Observations	Deviations		Standard deviation	Maximum	Minimum	Range	Overshoot		Undershoot	
		Average	Median					Number	Percent	Number	Percent
0.5	175	0.80	0.69	2.95	13.17	-10.44	23.61	115	65.71	60	34.29
1.0	169	0.69	0.81	1.82	8.98	-4.69	13.67	117	69.23	52	30.77
1.5	163	0.58	0.59	1.58	7.27	-3.65	10.91	100	61.35	63	38.65
2.0	157	0.47	0.42	1.33	5.11	-3.00	8.11	99	63.06	58	36.94
2.5	151	0.38	0.23	1.36	5.89	-2.66	8.55	85	56.29	66	43.71
3.0	145	0.31	0.06	1.19	4.94	-1.65	6.59	76	52.41	69	47.59
3.5	139	0.24	-0.01	1.17	4.79	-1.68	6.48	68	48.92	71	51.08
4.0	133	0.22	-0.14	1.07	4.26	-1.37	5.63	58	43.61	75	56.39
4.5	127	0.18	-0.03	1.06	4.18	-1.39	5.57	60	47.24	67	52.76
5.0	121	0.10	-0.13	0.92	3.69	-1.20	4.90	51	42.15	70	57.85
5.5	115	0.05	-0.13	0.84	3.68	-1.28	4.96	47	40.87	68	59.13
6.0	109	-0.01	-0.13	0.68	2.98	-1.12	4.10	41	37.61	68	62.39
6.5	103	-0.06	-0.15	0.68	2.82	-1.50	4.32	39	37.86	64	62.14
7.0	97	-0.09	-0.16	0.59	2.27	-1.31	3.58	33	34.02	64	65.98
7.5	91	-0.16	-0.23	0.62	2.13	-1.54	3.66	28	30.77	63	69.23
8.0	85	-0.21	-0.28	0.61	1.75	-1.44	3.20	25	29.41	60	70.59
8.5	79	-0.27	-0.33	0.62	1.56	-1.51	3.07	24	30.38	55	69.62
9.0	73	-0.38	-0.50	0.58	1.30	-1.46	2.76	18	24.66	55	75.34
9.5	67	-0.49	-0.56	0.50	0.66	-1.50	2.15	13	19.40	54	80.60
10.0	61	-0.47	-0.57	0.43	0.68	-1.04	1.71	9	14.75	52	85.25
10.5	55	-0.48	-0.57	0.43	0.51	-1.09	1.61	9	16.36	46	83.64
11.0	49	-0.50	-0.60	0.42	0.59	-1.06	1.65	7	14.29	42	85.71
11.5	43	-0.54	-0.66	0.40	0.35	-1.08	1.43	6	13.95	37	86.05
12.0	37	-0.53	-0.64	0.36	0.35	-1.01	1.36	4	10.81	33	89.19
12.5	31	-0.53	-0.61	0.36	0.17	-1.07	1.24	4	12.90	27	87.10
13.0	25	-0.51	-0.58	0.36	0.21	-1.05	1.26	3	12.00	22	88.00
13.5	19	-0.45	-0.44	0.37	0.12	-1.08	1.21	2	10.53	17	89.47
14.0	13	-0.28	-0.22	0.27	0.14	-0.61	0.75	2	15.38	11	84.62
14.5	7	-0.09	-0.16	0.13	0.09	-0.25	0.34	2	28.57	5	71.43
15.0	1	-0.19	-0.19	—	-0.19	-0.19	0.00	0	0.00	1	100.00

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

How well did inflation expectations based on TBI rates estimate CPI-U inflation before, during, and after the Great Recession?

This section examines the discrepancy between expected and actual inflation by calculating the difference between the TBI rate for a specific maturity horizon and its respective annualized CPI-U rate of change. Given the incremental nature of the data, this difference is calculated for every month from July 2003 to January 2018 for the 6-month maturity horizon (6-month TBI rates are available for matching to their respective annualized CPI-U inflation rates for all months between July 2003 and January 2018), for every month from January 2004 to January

2018 for the 1-year maturity horizon (1-year TBI rates are available for matching for all months between January 2004 and January 2018), and so on. Although the number of observations decreases with each maturity horizon, there are still 30 or more observations available for maturity horizons of 12.5 or fewer years. Given this framework, three main results follow.

Result 1: Average and median deviations never exceed 80 basis points

The analysis reveals that the average deviations between TBI rates and annualized CPI-U inflation rates never exceed 80 basis points for any 6-month increment (1 basis point equals one-hundredth of a percent, so 100 basis points equal 1 percent), and for maturity horizons of 2 or more years, they never exceed 55 basis points. The median deviations never exceed 70 basis points (except at the 1-year horizon). (See table 1 and figures 3 and 4.) If we take the average or the median of the average deviations per horizon, the TBI rate virtually equals the annualized CPI-U inflation rate (the average of averages per horizon is –7 basis points, and the median of averages per horizon is –13 basis points), and the same holds for the average or the median of the median deviations per horizon (the average of medians is –17 basis points, and the median of medians is –19 basis points).

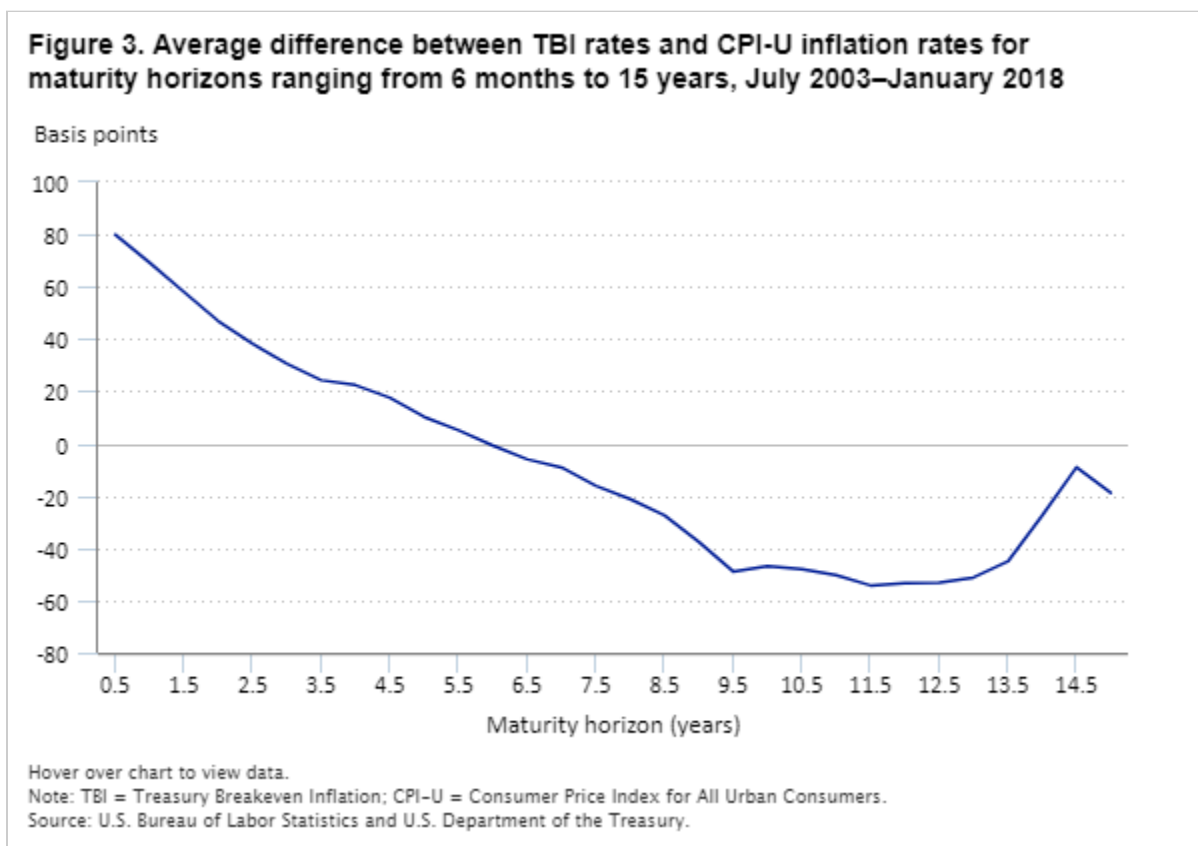
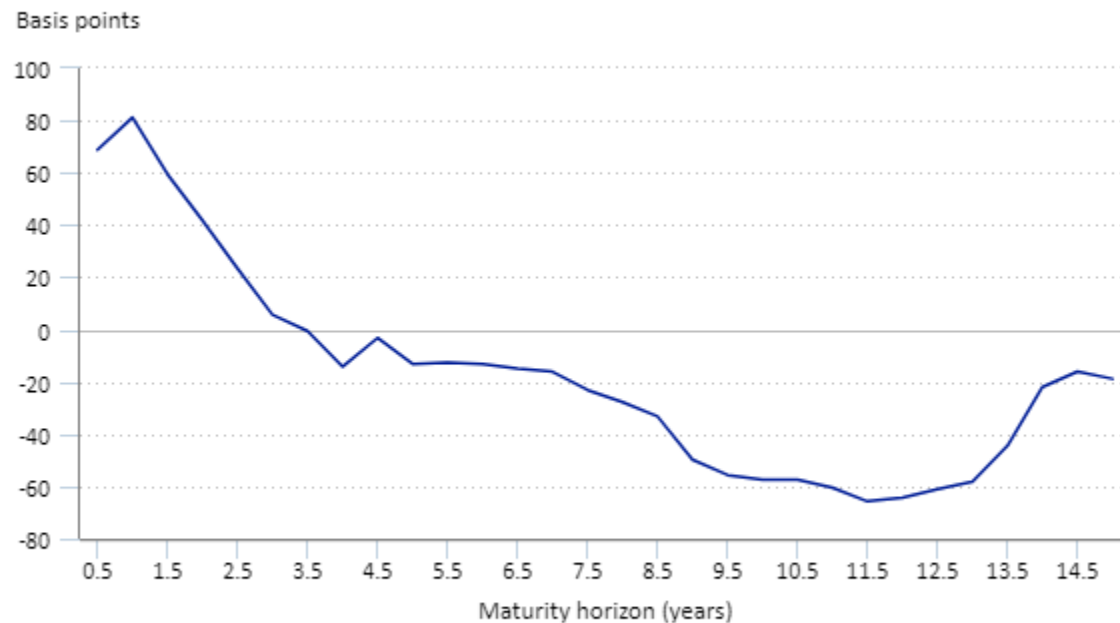


Figure 4. Median difference between TBI rates and CPI-U inflation rates for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018



Hover over chart to view data.

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

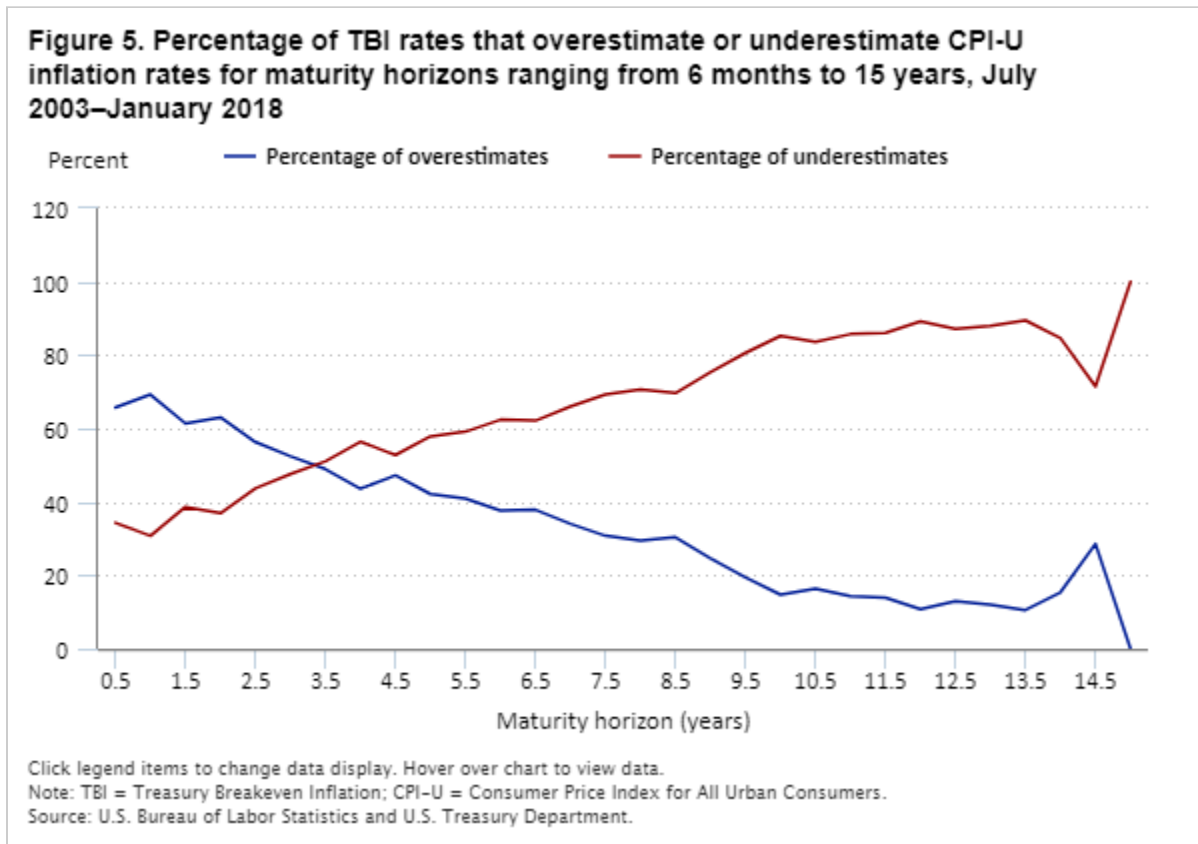
Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

Result 2: TBI expectations overshoot realized inflation for short maturity horizons and undershoot it for long maturity horizons

The percentage of TBI rates that exceed actual annualized inflation decreases as the maturity horizon increases. Of the 175 observations available at the 6-month maturity horizon, 66 percent of TBI rates overshoot the matching CPI-U inflation rates. At the 2-year maturity horizon, 63 percent of TBI rates overshoot the matching CPI-U inflation rates. For maturity horizons with 30 or more observations, this percentage steadily decreases until the 13-year maturity horizon, when 12 percent of TBI rates (out of 25 observations) overshoot the matching annualized CPI-U inflation rates. On the flip side, the percentage of TBI rates that undershoot actual inflation steadily rises as the maturity horizon increases. In both cases, however, the degree to which TBI rates overestimate or underestimate actual annualized inflation is small, as evidenced by the average and median deviations cited above. Given that TBI expectations tend to overshoot actual inflation in the short term and undershoot it in the long term, it is likely that a liquidity premium was embedded in the TIPS yield in the early years of TIPS issuance. TBI rates for longer horizons are available for matching to their respective annualized CPI-U inflation rates only for those early years.

As noted earlier, however, the TBI data begin in January 2003, 6 years after the first issuance of inflation-indexed debt. Given that the TIPS market was more established by then, it was more liquid, causing the liquidity premium to be smaller. Indeed, for maturities ranging from 3.5 years (the first horizon at which more than 50 percent of the observed TBI rates undershoot actual inflation) to 12.5 years (the horizon after which the number of observations falls below 30), the average deviation between expected and realized inflation never exceeds 54 basis points and the median deviation never exceeds 66 basis points. (Moreover, over this time span, the average of averages across all horizons is -21 basis points and the average of medians is -34 basis points.) Because observations at

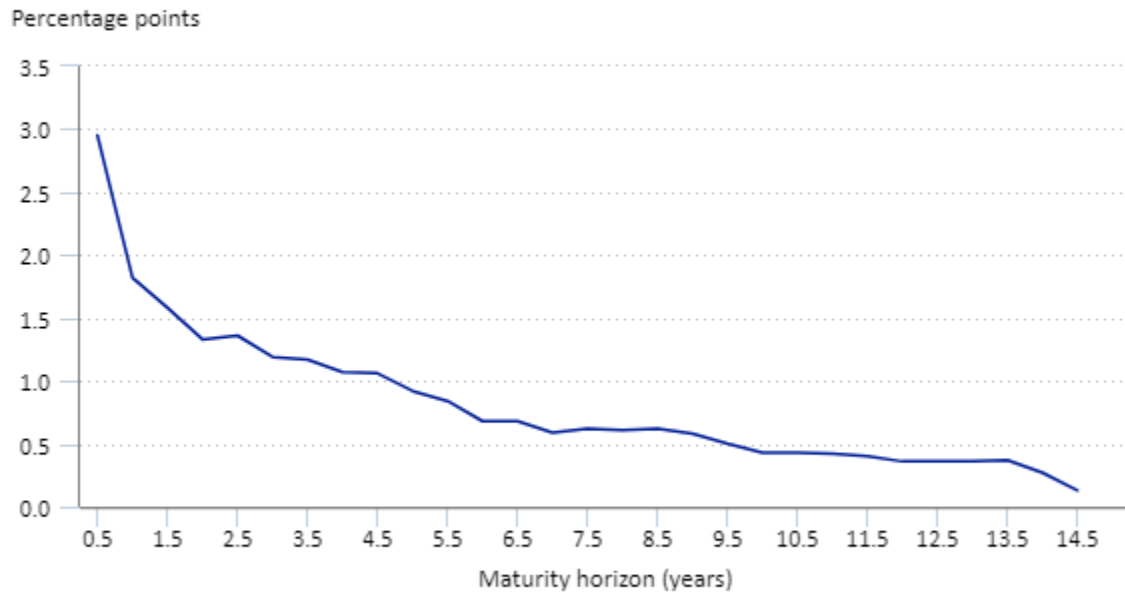
the 12.5-year maturity horizon begin in July 2015 and end in January 2018, they are based on TBI rates calculated for January 2003 through July 2004. (See figure 5.)



Result 3: The dispersion of differences between inflation expectations and reality decreases as the maturity horizon increases

While the average and median differences between TBI rates and actual annualized inflation rates are small, the dispersion of those differences decreases as the maturity horizon increases. This result holds when one measures dispersion by standard deviation, range, and maximum and minimum differences. The standard deviation of the differences between expected and realized inflation decreases from a peak of 2.95 at the 0.5-year horizon (175 observations) to 0.13 at the 14.5-year horizon (7 observations). (With only one observation at the 15-year horizon, the standard deviation is not defined.) The range of differences decreases from a peak of 23.61 at the 0.5-year horizon to 0.34 at the 14.5-year horizon. (With only one observation at the 15-year horizon, the maximum and minimum differences are the same.) The largest maximum difference between expected and realized inflation (13.17) is observed at the 0.5-year horizon, and the smallest maximum difference (0.09) is observed at the 14.5-year horizon. The largest minimum difference (−10.44) is observed at the 0.5-year horizon, and the smallest minimum difference (−0.25) is observed at the 14.5-year horizon. (See figures 6, 7, 8, and 9.)

Figure 6. Standard deviation of differences between TBI rates and CPI-U inflation rates for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018

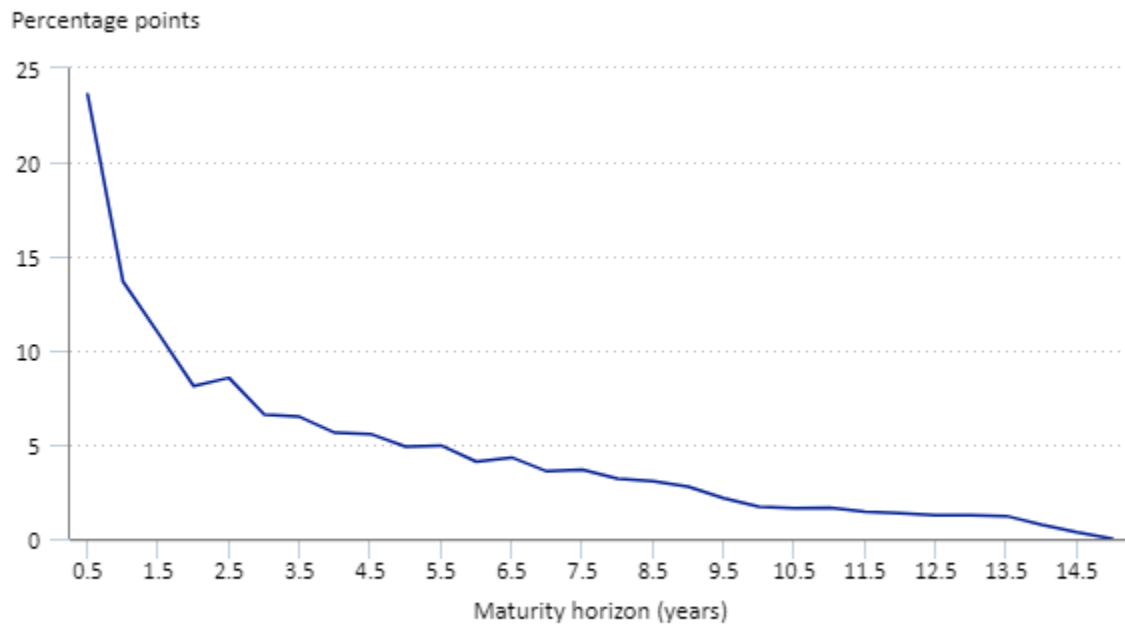


Hover over chart to view data.

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

Figure 7. Range of differences between TBI rates and CPI-U inflation rates for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018

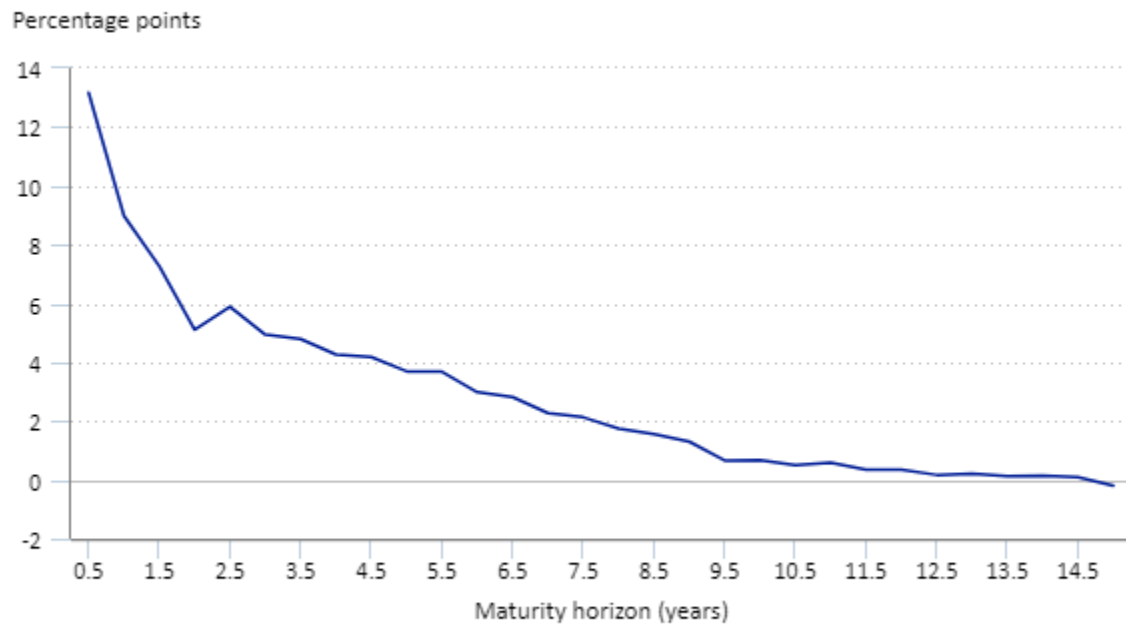


Hover over chart to view data.

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

Figure 8. Maximum difference between TBI rates and CPI-U inflation rates for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018

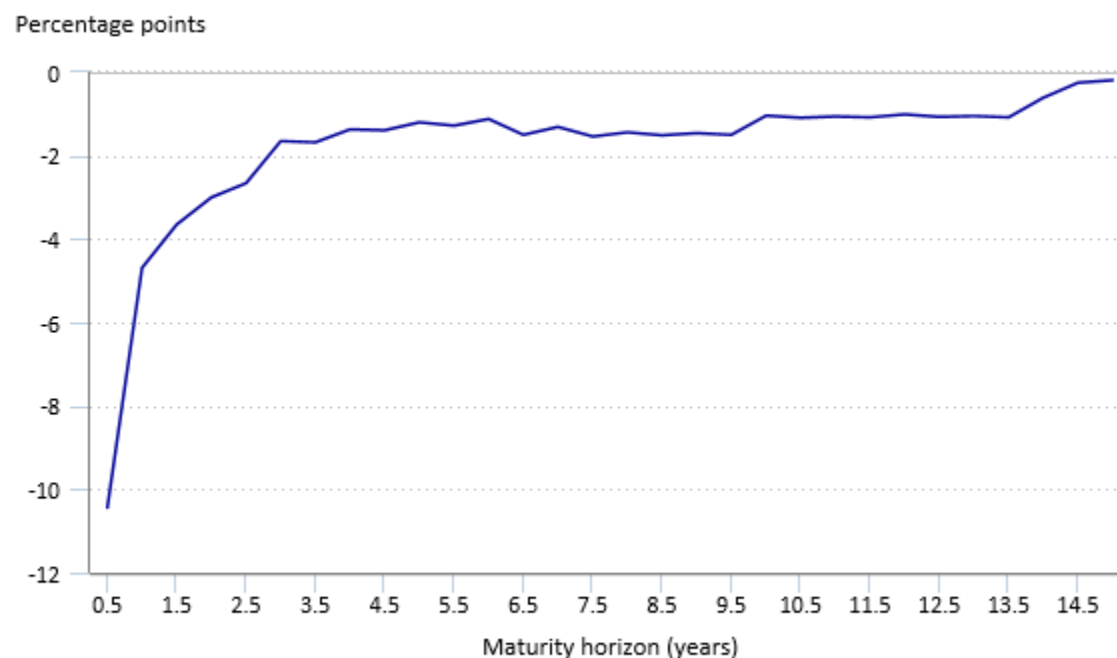


Hover over chart to view data.

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

Figure 9. Minimum difference between TBI rates and CPI-U inflation rates for maturity horizons ranging from 6 months to 15 years, July 2003–January 2018



Hover over chart to view data.

Note: TBI = Treasury Breakeven Inflation; CPI-U = Consumer Price Index for All Urban Consumers.

Source: U.S. Bureau of Labor Statistics and U.S. Department of the Treasury.

Conclusion

This article finds that TBI rates reasonably approximated inflation reality before, during, and after the Great Recession. In the analysis, the average and median deviations between TBI rates and their respective annualized CPI-U inflation rates never exceed 81 basis points, although a liquidity premium embedded in the TIPS yield probably explains why inflation expectations undershoot actual inflation for longer maturity horizons. Moreover, the dispersion of deviations, as measured by standard deviation and range, decreases as the maturity horizon increases.

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NOTES

¹ Treasury Inflation-Protected Securities (TIPS) would subsequently be renamed Treasury Inflation-Indexed Securities, but TIPS remains in the market lexicon.

² For a TIPS timeline, see https://www.treasurydirect.gov/indiv/research/history/histtime/histtime_tips.htm. See also https://www.treasurydirect.gov/indiv/research/indepth/tips/res_tips.htm.

³ John Y. Campbell and Robert J. Shiller, "A scorecard for indexed government debt" (National Bureau of Economic Research, January 1996), <https://www.nber.org/chapters/c11029.pdf>.

⁴ Caitlin Zaloom, "How to read the future: the yield curve, affect, and financial prediction," *Public Culture*, no. 21, no. 2, spring 2009, pp. 245–268, <https://as.nyu.edu/content/dam/nyu-as/faculty/documents/zaloom.howtoreadfuture.pdf>.

⁵ See Irving Fisher, "Appreciation and interest," *Publications of the American Economic Association*, vol. XI, no. 4, August 1896, pp. 331–442, <https://www.policonomics.com/wp-content/uploads/Appreciation-and-Interest.pdf>; Fisher, *The theory of interest, as determined by impatience to spend income and opportunity to invest it* (New York: The Macmillan Company, 1930), http://files.libertyfund.org/files/1416/Fisher_0219.pdf; and Fisher, *The rate of interest: its nature, determination, and relation to economic phenomena* (New York: The Macmillan Company, 1907), <https://socialsciences.mcmaster.ca/econ/ugcm/3ll3/fisher/RateofInterest.pdf>.

⁶ For more information on the derivation of the TBI curve, see James A. Girola, "The Treasury real yield curve and breakeven inflation," PowerPoint presentation (U.S. Department of the Treasury, July 21, 2015), https://www.treasury.gov/resource-center/economic-policy/corp-bond-yield/Documents/tii_may2015.pdf.

⁷ See *ibid.*, slides 33 and 34.

⁸ Olesya V. Grishchenko and Jing-Zhi Huang, "The inflation risk premium: evidence from the TIPS market," *The Journal of Fixed Income*, vol. 22, no. 4, spring 2013, pp. 5–30.

⁹ Geert Bekaert and Xiaozheng Wang, "Inflation risk and the inflation risk premium," *Economic Policy*, vol. 25, no. 64, October 2010, pp. 755–806, https://www0.gsb.columbia.edu/faculty/gbekaert/papers/inflation_risk.pdf.

¹⁰ Examples of papers attempting to estimate the liquidity premium include Charles T. Carlstrom and Timothy S. Fuerst, "Expected inflation and TIPS" (Federal Reserve Bank of Cleveland, November 1, 2004), <https://www.clevelandfed.org/newsroom-and-events/publications/economic-commentary/economic-commentary-archives/2004-economic-commentaries/ec-20041101-expected-inflation-and-tips.aspx>; Pu Shen, "Liquidity risk premia and breakeven inflation rates," *Economic Review* (Federal Reserve Bank of Kansas City, 2nd quarter 2006), <https://www.kansascityfed.org/QYdNZ/Publicat/econrev/PDF/2Q06Shen.pdf>; Jens Christensen and James Gillan, "TIPS liquidity, breakeven inflation, and inflation expectations," *Economic Letter* (Federal Reserve Bank of San Francisco, June 20, 2011), <https://www.frbsf.org/economic-research/publications/economic-letter/2011/june/tips-liquidity-breakeven-inflation-expectations/>; Michael Abrahams, Tobias Adrian, Richard K. Crump, and Emanuel Moench, "Decomposing real and nominal yield curves," Staff Report 570 (Federal Reserve Bank of New York, September 2012, revised February 2015), https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr570.pdf; Carolin E. Pflueger and Luis M. Viceira, "An empirical decomposition of risk and liquidity in nominal and inflation-indexed government bonds" (Boston, MA: Harvard Business School, March 2012); Stefania D'Amico, Don H. Kim, and Min Wei, "Tips from TIPS: the informational content of Treasury Inflation-Protected Security prices," *Finance and*

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11 Andrew Ang, Geert Bekaert, and Min Wei, “Do macro variables, asset markets, or surveys forecast inflation better?” *Journal of Monetary Economics*, vol. 54, no. 4, May 2007, pp. 1163–1212.

12 For more information on these curves, see James A. Girola, “The yield curve for Treasury nominal coupon issues,” PowerPoint presentation (U.S. Department of the Treasury, May 16, 2014), https://www.treasury.gov/resource-center/economic-policy/corp-bond-yield/Documents/trp_apr2014.pdf; and Girola, “The Treasury real yield curve and breakeven inflation.”

13 As explained by Girola (“The Treasury real yield curve and breakeven inflation,” slide 15), “This is because no noticeable on-the-run effects appear in the TIPS market. Moreover, the number of TIPS is so small that it would not be possible to sort out statistically any on-the-run effects. Therefore, the TNC regression variables for on-the-run and first off-the-run are omitted from the TRC yield curve.”

14 TNC, TRC, and TBI data are available at <https://www.treasury.gov/resource-center/economic-policy/corp-bond-yield/Pages/TNC-YC.aspx>.

15 According to the Blended Retirement System policy of the Uniformed Services, “In computing the amount of the lump sum described in 7.a.(3)(c) of Attachment 1, the discounted present value will be determined in accordance with the rate that is an inflation-adjusted 7-year average of the Department of Treasury High-Quality Market (HQM) Corporate Bond Spot Rate Yield Curve at a 23-year maturity plus an adjustment factor of 4.28 percentage points. The inflation adjustment applied is the Department of Treasury ‘Breakeven Inflation Spot Rate Yield Curve.’” See attachment 2 in “Implementation of the Blended Retirement System,” memorandum (U.S. Department of Defense, January 27, 2017), <http://militarypay.defense.gov/Portals/3/Documents/BlendedRetirementDocuments/Combined%20BRS%20Policy%20Document.pdf?ver=2018-01-02-105828-370>.

16 For more information, see chapter 17, “The Consumer Price Index (updated 2-14-2018),” in *Handbook of methods* (U.S. Bureau of Labor Statistics, updated February 14, 2018), <https://www.bls.gov/opub/hom/pdf/cpi-20180214.pdf>.

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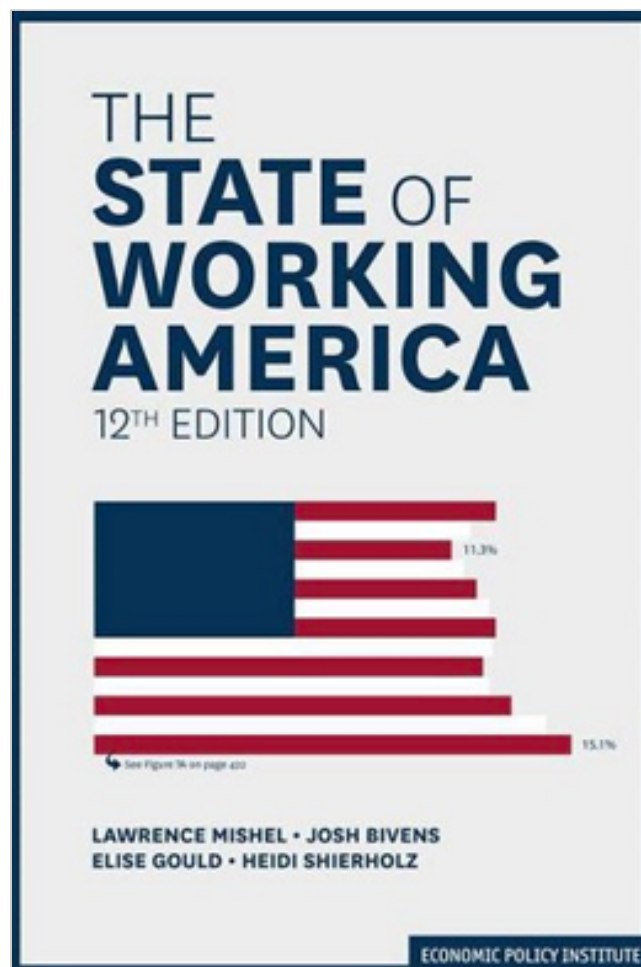
The American working class and the unfulfilled promise of policy-driven growth

The State of Working America, 12th ed. By Lawrence Mishel, Josh Bivens, Elise Gould, and Heidi Shierholz. Ithaca, New York: ILR Press, 2012, 505 pp., \$24.95 paperback.

For over two centuries, the United States has been called “the land of opportunity,” a place where individuals and families can live and prosper in freedom. This popular idea, frequently referred to as the American Dream, suggests that, in a country where success has historically seen no barriers, the joys of financial well-being are up for grabs for anyone willing to work hard. It is on this basis that many people still assume that the United States remains a land of opportunity. In *The State of Working America*, however, authors Lawrence Mishel, Josh Bivens, Elise Gould, and Heidi Shierholz argue that this assumption is no longer safe.

The book asks one seemingly simple question: “How well has the American economy worked to provide acceptable growth in living standards for most households?” According to the authors, this question cannot be answered simply by looking at output growth and inflation rates. Over the course of seven chapters, Mishel and colleagues dissect the American economy along multiple dimensions, examining what has happened to American households. Their verdict, summarized in the book’s opening chapter, is discomfoting—for most people, the economy has worked “not well at all.”

The authors’ analysis focuses on economic policy, income inequality, wage growth, social mobility, productivity, unemployment rates, wealth, and poverty. It also examines how race, education, gender, and other demographic characteristics influence these critical metrics. While the discussion is painstakingly detailed, it is surprisingly



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accessible, making even complex material easy to understand. The results of the analysis are often broken down by income quintiles, allowing the reader to see how each facet of the economy has affected different income groups in their pursuit of financial success and higher living standards. The data come primarily from the Current Population Survey (CPS) Annual Social and Economic Supplement, full monthly public data series, and Outgoing Rotation Group files. Information on household income, poverty, pensions, and healthcare benefits is derived from the CPS March supplement.

The authors' main conclusion is that, for the nearly 40 years between the late 1970s and the time of the book's publication, "economic policy has increasingly served the interests of those with the most wealth, income, and political power." This development is attributed to two key policy failures: allowing businesses to crush worker unionization efforts and failing to raise the minimum wage. Together, these setbacks have restricted the ability of most Americans to bargain for better working conditions and provided no counterbalance to inflation, which has reduced workers' purchasing power and squeezed those at the bottom of the income distribution.

But income inequality is only one of the many predicaments faced by American workers. The authors also show that, despite their best effort and determination, these workers have been unable to break class barriers and achieve upward mobility. In general, the class in which most Americans are born is likely the class in which they will remain for the duration of their lives. In the authors' grim assessment, "there is no evidence that mobility has increased to offset rising inequality, and in fact some research shows a decline."

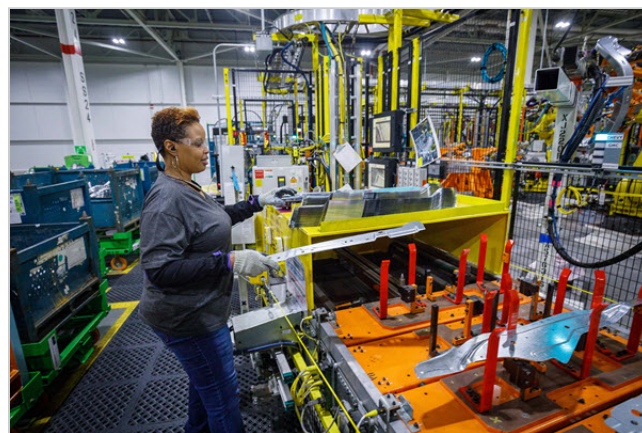
Unsurprisingly, the barriers faced by White, Black, and Hispanic workers are not the same. While many Americans might feel the burden of income inequality and plummeting purchasing power, some racial and ethnic groups have fared worse than others. Addressing the unemployment rate for Black workers, for example, the authors write that "African Americans have essentially been living through a perpetual, slow-motion recession." Between 1979 and 2012, the average unemployment rate for Black workers was over 12 percent, and that for Hispanics was only slightly lower. These rates far exceeded those for White workers.

How Americans experience life also differs by gender. Given much noise in the news media, it can be hard to understand what is happening to the gap between men and women. Offering a more indepth analysis, the authors reveal some troubling indications. Although wages for women have increased—and there has been some progress on bridging the gender pay gap—all is not as well as it appears. Women still lag behind men when it comes to pension and healthcare benefits. Moreover, the authors find that the narrowing of the gender pay gap has been due mostly to men losing benefits and wages, rather than to women gaining ground.

The authors do a wonderful job at showing how policy inaction has been at the core of the unique challenges faced by people of different income levels, genders, and races. It is new policy ideas, rather than existing policy, that could greatly change the odds of success for most American households. There will always be "winners" and "losers" when it comes to economic policymaking, but the policies America enacts should target and help the right people. *The State of Working America* helps lead the way to opening discussions about how economic and social policy can create a level playing field for those who have spent so many decades falling behind.

The relative weakness in earnings of production workers in manufacturing, 1990–2018

Although manufacturing industries had a reputation for stable, well-paying jobs for much of the 20th century, shifts within the industry in the last several decades have considerably altered that picture. Since 1990, average hourly earnings trends in the various manufacturing industries have been disparate, with a few industries showing strong growth but many others showing growth rates that are lower than those of the total private sector. In fact, average hourly earnings of production and nonsupervisory workers in the total private sector have recently surpassed those of their counterparts in the relatively high-paying durable goods portion of manufacturing.



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Until recently, the U.S. economy has been defined by its manufacturing capabilities. Manufacturing had historically been the dominant sector in terms of value of the output produced, and the pay earned by production workers in manufacturing has long served as a marker for what constitutes middle-class status in the United States.^[1] But that status has changed over the last several decades, as the service-providing sector has grown to replace manufacturing as the dominant driver of employment growth. Despite some recent job gains, most manufacturing industries have shed workers at high rates since 1990, and many employers in the industry have substantially reduced the growth rate of their workers' earnings—or, in some cases, they have even reduced the wages paid to their remaining workers. As a result, average hourly earnings of production workers in manufacturing have fallen below those of total private sector workers.^[2] (See table 1.)

Table 1. Employment and average hourly earnings of production and nonsupervisory workers in total private and production workers in manufacturing, not seasonally adjusted, 1990 and 2018 annual averages

Industry	Employment	Percent of total private employment	Average hourly earnings (AHE)	Percent of total private AHE
1990				
Total Private	73,721,000	—	\$10.20	—
Manufacturing	12,669,000	17.2%	10.78	105.7%
Durable goods	7,397,000	10.0	11.40	111.8
Nondurable goods	5,272,000	7.2	9.87	96.8
2018				
Total Private	104,319,000	—	22.71	—
Manufacturing	8,899,000	8.5	21.54	94.8
Durable goods	5,463,000	5.2	22.51	99.1
Nondurable goods	3,436,000	3.3	19.96	87.9

Note: Dash = not applicable.

Source: U.S. Bureau of Labor Statistics.

In 1990, average hourly earnings of production workers in manufacturing (\$10.78) were about 6 percent greater than those of production or nonsupervisory workers in the total private sector (\$10.20). By 2018, however, manufacturing workers were earning approximately 5 percent less (\$21.54) than their total private sector counterparts (\$22.71). Over this same period, manufacturing production workers' share of total private sector employment fell by half, from 17.2 percent in 1990 to 8.5 percent in 2018. This article analyzes earnings trends in manufacturing since 1990 and examines some of the component industries that have driven these broader trends. All earnings data cited in this article are from the U.S. Bureau of Labor Statistics (BLS) Current Employment Statistics (CES) survey and exclude all forms of benefits and other compensation employees may receive (other than their pay).^[3]

Earnings of production and nonsupervisory employees

Production workers made up 70.1 percent of all manufacturing workers in 2018—down slightly from 71.6 percent in 1990.^[4] Within manufacturing, production workers make up an array of workers, including “working supervisors or group leaders who may be ‘in charge’ of some employees, but whose supervisory functions are only incidental to their regular work.”^[5] In this article, earnings for production workers in manufacturing are compared with earnings for construction workers in construction, production workers in mining and logging, and nonsupervisory workers in private service-providing industries. Although the definitions of these worker categories differ slightly, the workers are analytically comparable and together constitute all production and nonsupervisory employees in the total private sector.^[6]

Long-term analysis is particularly well-suited for earnings of production and nonsupervisory employees because these data are available over a much longer period than earnings for all employees. Although earnings measures for all employees within manufacturing are currently available, the data series only goes back to March 2006.

Average hourly earnings of production and nonsupervisory employees by industry

In 1990, average hourly earnings for production workers in manufacturing as a whole were above those of production and nonsupervisory employees in the total private sector, but this was not true across all component industries within manufacturing. Production workers in durable goods manufacturing earned 12 percent more, on average, than their counterparts in the total private sector, while workers in nondurables goods manufacturing earned 3 percent less than comparable workers in the total private sector. Moreover, the spread in earnings among workers in the various manufacturing industries was quite large, with those in petroleum and coal products earning \$17.00 per hour, on average, and apparel workers earning \$6.22 per hour, a difference of \$10.78. (See table 2.)

Table 2. Employment and average hourly earnings of production and nonsupervisory workers, total private, manufacturing, and selected other industries, not seasonally adjusted, 1990 and 2018 annual averages

Industry	1990		2018		Annualized percent change, 1990 to 2018	
	Employment	Average hourly earnings	Employment	Average hourly earnings	Employment	Average hourly earnings
Total private	73,721,000	\$10.20	104,319,000	\$22.71	1.2%	2.9%
Manufacturing	12,669,000	10.78	8,899,000	21.54	-1.3	2.5
Durable goods	7,397,000	11.40	5,463,000	22.51	-1.1	2.5
Wood products	451,500	8.82	319,000	17.88	-1.2	2.6
Nonmetallic mineral products	413,200	11.11	309,900	21.41	-1.0	2.4
Primary metals	525,100	12.97	293,600	23.31	-2.1	2.1
Fabricated metal products	1,190,100	10.64	1,084,900	20.55	-0.3	2.4
Machinery	938,900	11.73	718,200	23.06	-1.0	2.4
Computer and electronic parts	980,200	10.89	613,100	25.09	-1.7	3.0
Electrical equipment and appliances	465,200	10.00	260,500	20.59	-2.0	2.6
Transportation equipment	1,473,400	14.44	1,188,300	26.32	-0.8	2.2
Motor vehicles and parts	869,500	15.00	778,700	22.77	-0.4	1.5
Furniture and related products	475,200	8.53	291,500	17.86	-1.7	2.7
Miscellaneous durable goods	484,200	8.87	384,100	19.14	-0.8	2.8
Nondurable goods	5,272,000	9.87	3,436,000	19.96	-1.5	2.5
Food manufacturing	1,165,000	9.04	1,271,500	17.49	0.3	2.4
Textile mills	417,900	8.17	87,700	16.46	-5.4	2.5
Textile product mills	194,900	7.37	85,600	15.31	-2.9	2.6
Apparel	805,200	6.22	80,900	15.32	-7.9	3.3
Paper and paper products	493,200	12.06	276,100	21.88	-2.1	2.2
Printing and related support activities	597,600	11.11	294,900	18.74	-2.5	1.9
Petroleum and coal products	97,500	17.00	77,400	40.32	-0.8	3.1

See footnotes at end of table.

Table 2. Employment and average hourly earnings of production and nonsupervisory workers, total private, manufacturing, and selected other industries, not seasonally adjusted, 1990 and 2018 annual averages

Industry	1990		2018		Annualized percent change, 1990 to 2018	
	Employment	Average hourly earnings	Employment	Average hourly earnings	Employment	Average hourly earnings
Chemicals	620,300	12.85	547,700	25.46	-0.4	2.5
Plastics and rubber products	646,700	9.76	548,100	18.49	-0.6	2.3
Miscellaneous nondurable goods	233,800	10.28	165,700	20.11	-1.2	2.4
Mining and logging	538,000	13.40	544,000	28.30	0.0	2.7
Construction	4,115,000	13.42	5,438,000	27.74	1.0	2.6
Wholesale trade	4,167,500	11.55	4,698,000	25.18	0.4	2.8
Retail trade	11,311,000	7.71	13,529,200	15.91	0.6	2.6
Transportation and warehousing	2,943,200	12.50	4,721,000	21.84	1.7	2.0
Utilities	584,900	16.14	444,700	36.77	-1.0	3.0
Information	1,866,000	13.40	2,278,000	31.93	0.7	3.1
Financial activities	4,973,000	9.98	6,637,000	26.94	1.0	3.6
Credit intermediation and related services	1,808,100	9.06	2,015,000	23.72	0.4	3.5
Real estate	849,100	8.71	1,255,800	21.69	1.4	3.3
Rental and leasing	415,000	8.49	458,400	20.45	0.4	3.2
Professional and business services	8,915,000	11.15	17,123,000	26.81	2.4	3.2
Professional and technical services	3,499,800	14.00	7,255,900	35.41	2.6	3.4
Management of companies and enterprises	1,279,900	11.11	1,540,200	28.86	0.7	3.5
Administrative and waste services	4,135,100	8.48	8,326,700	18.32	2.5	2.8
Education and health services	9,784,000	9.98	20,788,000	23.65	2.7	3.1
Healthcare	7,344,400	10.42	14,121,200	25.92	2.4	3.3
Social assistance	966,900	6.99	3,478,400	14.84	4.7	2.7
Leisure and hospitality	8,299,000	6.02	14,382,000	13.87	2.0	3.0
Other services	3,555,000	9.08	4,839,000	20.78	1.1	3.0

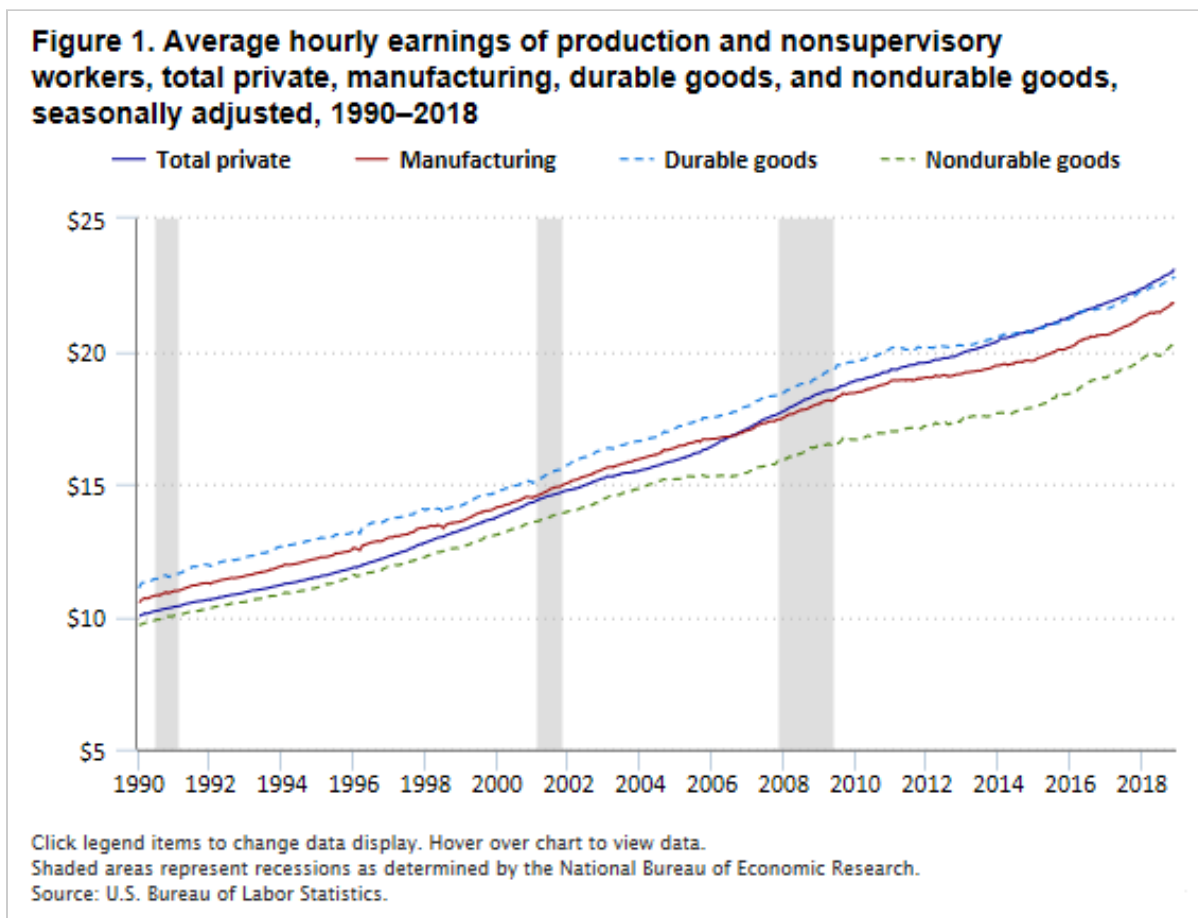
Source: U.S. Bureau of Labor Statistics.

In 1990, workers in seven nonmanufacturing sectors earned more per hour than workers in manufacturing, while workers in five nonmanufacturing sectors—leisure and hospitality, retail trade, financial activities, education and health services, and other services—earned less than workers in manufacturing. By 2018, hourly earnings in manufacturing exceeded those of only three other major industries: leisure and hospitality, retail trade, and other services. In that same year, workers in financial activities and in private education and health services earned more than manufacturing workers. Furthermore, the spread in hourly earnings among the various manufacturing

industries remained large nearly three decades later, with workers in petroleum and coal products earning \$40.32 per hour, \$25 more than workers in textile product mills.

Average hourly earnings of production and nonsupervisory workers, total private and manufacturing

On an annualized basis, hourly earnings of manufacturing workers rose by 2.5 percent per year from 1990 to 2018, while private sector average earnings rose by 2.9 percent. As a result of this disparity in the rate of growth over the period, the level of the two data series converged in July 2006. Since then, earnings in manufacturing have remained below those of private sector workers, on average. Several factors contributed to this change. First, as previously noted, hourly earnings in manufacturing rose more slowly over the period than total private hourly earnings. Second, hourly earnings are weighted by employment. Thus, because workers in durable goods industries generally earn more per hour than those in nondurable goods, the loss of 98,000 more jobs in durable goods than in nondurable goods over the period pulled the hourly earnings average for manufacturing down further.^[7] In fact, hourly earnings of workers in the durable goods industries fell below those of total private sector workers in 2015, and they have remained below them since then. (See figure 1.)



Average weekly earnings of manufacturing workers and total private workers

In January 1990, average weekly earnings of manufacturing workers were 24.7 percent higher than those of private sector workers, and by the close of 2018, that figure had fallen to 18.4 percent.[\[8\]](#) (See table 3.)

Table 3. Average weekly earnings in manufacturing and percent difference from total private, annual averages, not seasonally adjusted

Industry	1990	Percent difference	2018	Percent difference
Total private	\$349.63	—	\$767.08	—
Manufacturing	436.13	24.7%	908.08	18.4%
Durable goods	468.43	34.0	956.88	24.7
Nondurable goods	390.65	11.7	830.08	8.2

Note: Dash = not applicable.

Source: U.S. Bureau of Labor Statistics

Over the 1990–2018 period, weekly earnings of total private production and nonsupervisory workers grew at an annualized rate of 2.8 percent per year, while weekly earnings of manufacturing workers grew by just 2.7 percent over that period. Thus, although *hourly* earnings in manufacturing have been eclipsed by hourly earnings in total private, *weekly* earnings in manufacturing continue to be substantially higher than weekly earnings in the private sector as a whole. Average weekly earnings for any industry are the product of average hourly earnings and average weekly hours, so any differences in the trends over time between various industries' average weekly earnings and average hourly earnings stem from changes in average weekly hours.[\[9\]](#)

Average weekly hours of production workers in manufacturing increased from 40.5 to 42.2 over the 1990–2018 period, while weekly hours in the total private sector decreased from 34.3 to 33.8. (See table 4.)

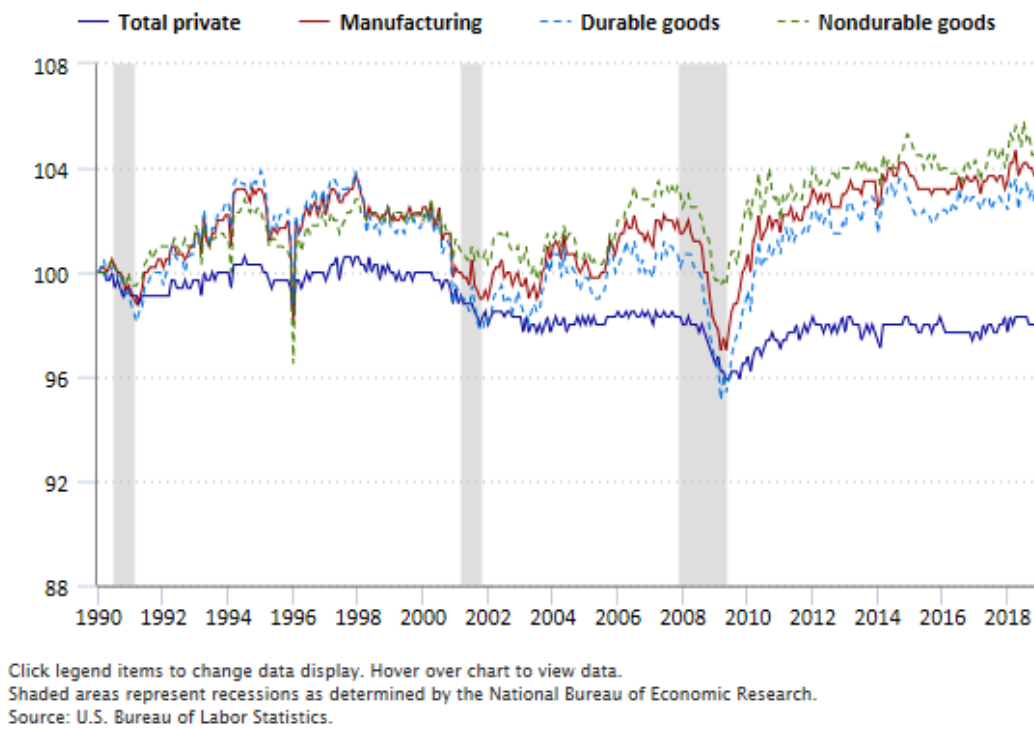
Table 4. Average weekly hours of production workers in manufacturing and production and nonsupervisory workers in total private, not seasonally adjusted, 1990 and 2018 annual averages

Industry	1990	2018
Total Private	34.3	33.8
Manufacturing	40.5	42.2
Durable Goods	41.1	42.5
Nondurable Goods	39.6	41.6

Source: U.S. Bureau of Labor Statistics.

Whereas more hiring in industries with lower average weekly hours—such as retail trade and leisure and hospitality—has driven down the average workweek for production and nonsupervisory workers in the total private sector, weekly hours have increased in manufacturing, while employment has declined. In addition, compared with private sector hours, manufacturing hours show greater volatility from month to month. (See figure 2.)

Figure 2. Index of average weekly hours of production and nonsupervisory workers, seasonally adjusted, 1990–2018 (1990 = 100)



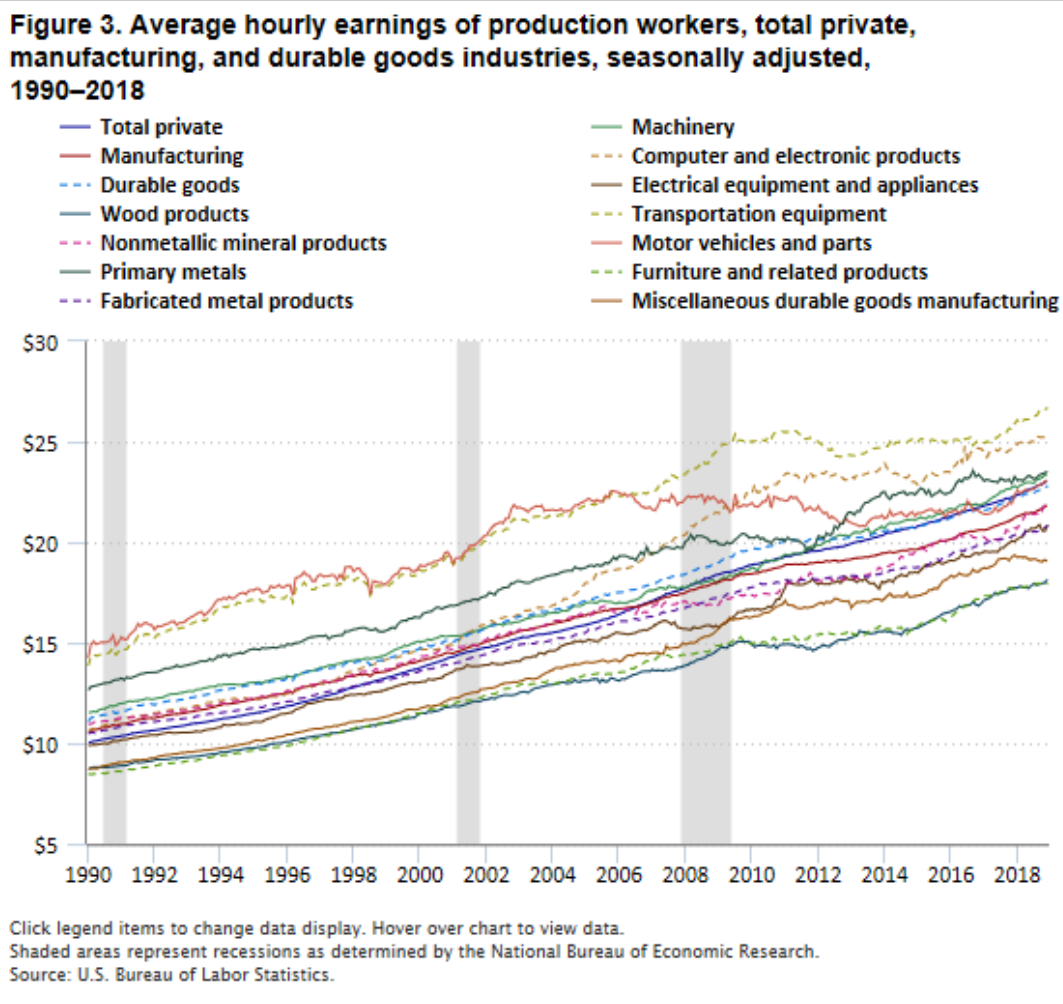
Average weekly hours in manufacturing have consistently been 6 to 8 hours higher than average weekly hours in total private. The longer workweek has kept weekly earnings in manufacturing above weekly earnings in total private, despite hourly earnings moving in the opposite direction—in 2018, average hourly earnings (\$22.71) in total private were slightly higher than average hourly earnings in manufacturing (\$21.54). Thus, workers in manufacturing have had to work more hours per week to make up for their relative weakness in hourly pay. In addition, employment has declined across virtually all of the component industries in manufacturing since 1990, and hourly earnings for many production workers in manufacturing have declined to below those of production and nonsupervisory workers in the total private sector. Nevertheless, the expanding workweek in manufacturing has kept average weekly earnings in the industry above average weekly earnings in the total private sector through 2018.

Durable goods and nondurable goods

Since 1990, hourly earnings in manufacturing's two major components, durable goods and nondurable goods, have risen more slowly than earnings in the total private sector. The relatively high average hourly earnings of production workers in durable goods manufacturing consistently kept hourly earnings higher in manufacturing than in the private sector as a whole until they converged in July 2006. Then, around 2011, hourly earnings growth in durable goods slowed to such an extent that they were relatively equal to those of the average private sector job for much of the period from 2014 to 2016. Since then, hourly earnings in durable goods have fallen slightly below

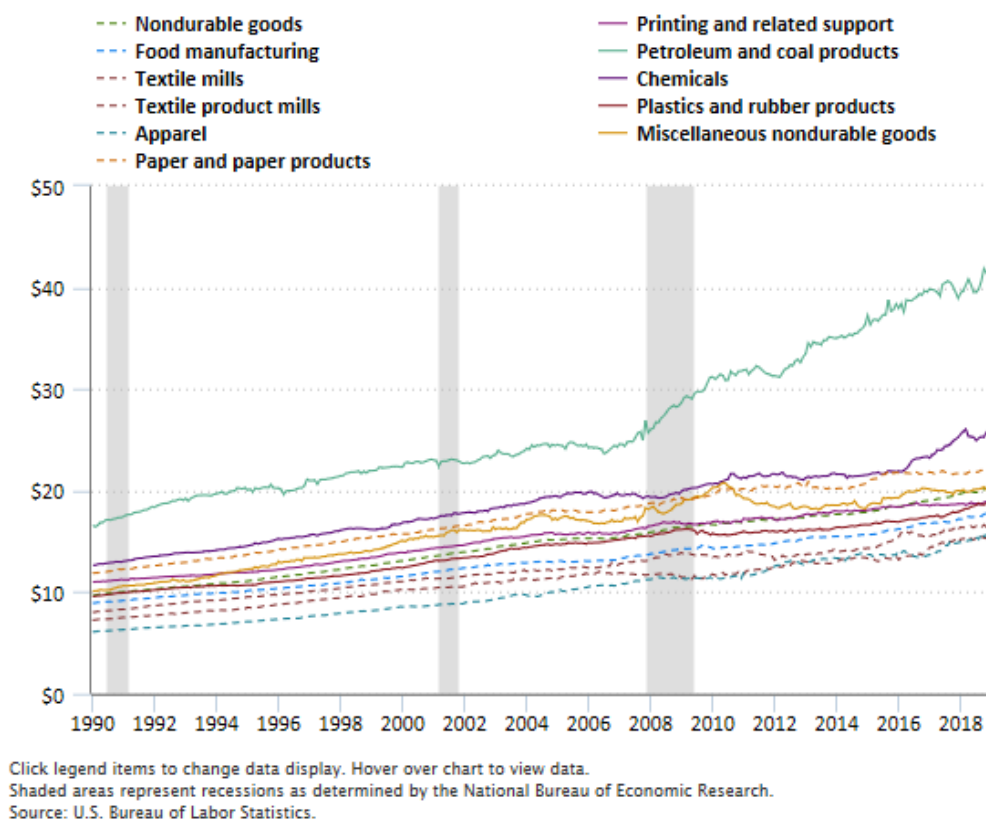
those of total private. In 1990, hourly earnings in durable goods were 12 percent higher than those in the private sector. By 2018, earnings in durable goods were 1 percent lower than earnings in the total private sector.

Hourly earnings in nondurable goods have consistently been below those in the total private sector, and they have grown more slowly as well. In 1990, hourly earnings in the total private sector were 3 percent higher than those in nondurable goods. By 2018, earnings in the total private sector were 14 percent higher than earnings in nondurable goods. Similarly, hourly earnings grew more slowly in nearly all industries within durable goods than in total private over the 1990–2018 period. The motor vehicles and parts industry stands out as having the slowest rate of growth in hourly earnings among all manufacturing industries, both durables and nondurables. Within durable goods, the largest rise and the most rapid growth rate in hourly earnings over the period occurred in computer and electronic products. (See figure 3.)



Hourly earnings in most nondurable goods industries in manufacturing also grew more slowly than those of the private sector. Printing and related activities experienced the weakest earnings growth, while two industries—apparel and petroleum and coal products—had above-average growth. (See figure 4.)

Figure 4. Average hourly earnings of production workers, nondurable goods, seasonally adjusted, 1990–2018



Motor vehicles and parts

As shown in table 2, hourly earnings in motor vehicles and parts, at \$15.00, were the highest of all durable goods industries in 1990. By 2018, that figure stood at \$22.77, which is a 1.5-percent annualized increase, the slowest growth rate of all industries in manufacturing over the 1990–2018 period. Meanwhile, total private hourly earnings grew by 2.9 percent per year over the same period, or nearly twice the rate of growth in motor vehicles and parts.

Although hourly earnings growth in the motor vehicles and parts industry was relatively slow over the period, average weekly hours in the industry increased by 10 percent. In fact, as of December 2018, motor vehicles and parts had the highest average weekly hours of all durable goods industries. As a result of the growth in the industry’s workweek, weekly earnings rose from \$621.58 in 1990 to \$1,017.37 in 2018, an annualized 1.8-percent growth rate, which was still slower than the 2.8-percent rate for the private sector as a whole.

The relatively small gains in hourly and weekly earnings over the period did not keep pace with inflation. As a result, for the motor vehicles and parts industry, both inflation-adjusted average hourly earnings and inflation-adjusted average weekly earnings declined over the 1990–2018 period. (See table 5.)

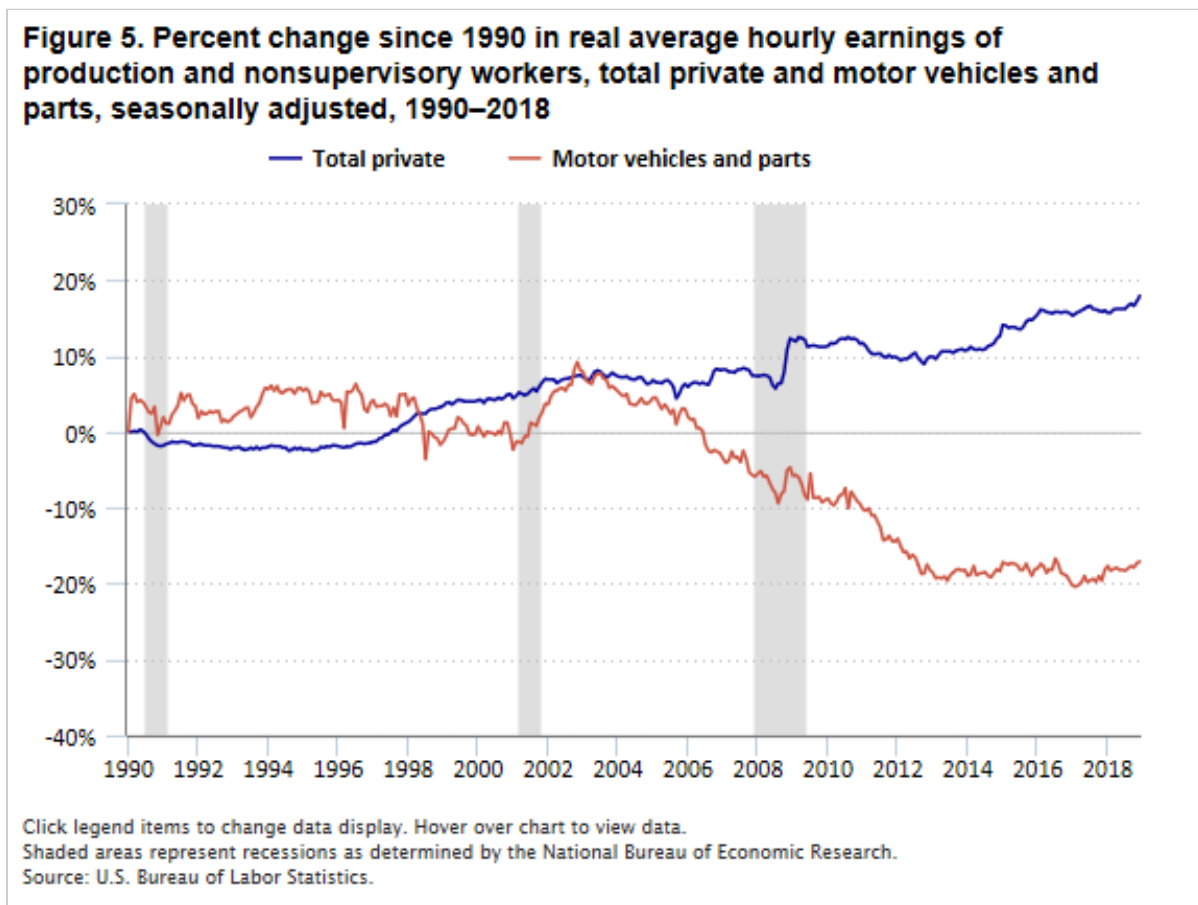
Table 5. Real average hourly earnings and real average weekly earnings of production and nonsupervisory workers, total private and motor vehicles and parts, annual averages, not seasonally adjusted, 1990–2018

Industry	Real hourly earnings			Real weekly earnings		
	1990	2018	Percent change	1990	2018	Percent change
Total private	\$7.91	\$9.26	17.1%	\$271.03	\$312.91	15.5%
Motor vehicles and parts	11.63	9.29	-20.1	481.84	415.01	-13.9

Note: Real earnings are earnings adjusted for inflation.

Source: U.S. Bureau of Labor Statistics

The decline in earnings represents a decline in the purchasing power of these workers. By contrast, inflation-adjusted hourly and weekly earnings in the total private sector increased over the period. (See figure 5.)



The driving force behind the stagnation in earnings was contract negotiations and concessions agreed to by the United Automobile Workers (UAW) and the automobile manufacturers. Negotiations between labor and management throughout much of the second half of the 20th century led to the establishment of pension plans and certain standard-of-living guarantees at Detroit's three major automobile manufacturers.

Nevertheless, recessionary pressures and competition from foreign automobile manufacturers began to exert downward pressure on earnings of workers in the industry during the 1990–2018 period.

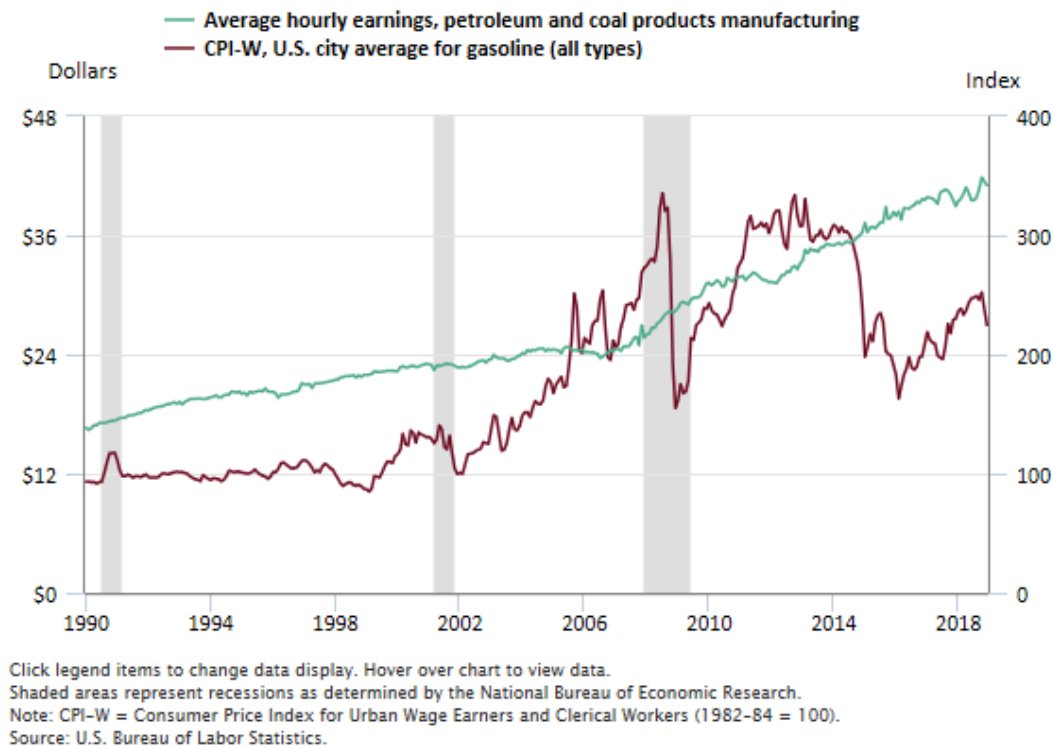
As sales of imported vehicles increased in the late 1970s and early 1980s, the dominance of domestic manufacturers in the U.S. automobile market declined. To address the impact of weak sales and to help ensure the survival of the industry, automakers and the UAW agreed to several concessions affecting autoworkers' pay:

- In 2007, a contract between the UAW and the three major U.S. automakers eliminated pay increases, while allowing signing bonuses, cost-of-living adjustments, and certain lump-sum bonus payments to continue. According to the Center for Automotive Research, the effect of these concessions by autoworkers was to lower average hourly labor costs from an estimated \$72 to \$78 per hour to approximately \$50 to \$58 per hour.[\[10\]](#)
- In 2009, following the hardships brought on by the Great Recession (December 2007 to June 2009), the 2007 agreement was modified to suspend the cost-of-living adjustments and certain bonus payments while tightening overtime and break-time allowances.[\[11\]](#)
- In 2011, the job banks for idled workers were eliminated and cost-of-living adjustments remained suspended.[\[12\]](#)

Petroleum and coal products

Production workers in petroleum and coal products manufacturing experienced the largest increase in hourly earnings among all manufacturing workers over the 1990–2018 period, with their hourly wages rising from \$17.00 to \$40.32. A driving force behind the strong earnings growth in the industry has been the price of oil and, by extension, gasoline. Figure 6 shows average hourly earnings of production workers in petroleum and coal products manufacturing over the 1990–2018 period, as well as the Consumer Price Index for Urban Wage Earners and Clerical Workers (CPI-W), U.S. city average for gasoline (all types). As the figure shows, for much of the period, earnings in the industry have increased with the rising price of gasoline. However, there tends to be a lag in this effect—the sharp increase in gasoline prices that occurred from 2002 to 2008 resulted in a steep increase in average hourly earnings from 2007 to 2011. As the price of gas oscillated substantially during the 2008–18 period (within a range well above pre-2002 levels), hourly earnings continued to increase steadily during the period from about 2012 to 2018.

Figure 6. Average hourly earnings of production workers in petroleum and coal products manufacturing and CPI-W, U.S. city average for gasoline (all types), 1990–2018, seasonally adjusted



Computer and electronic products

In 1990, average hourly earnings in computer and electronic products manufacturing were 4 percent below those of all durable goods industries combined. By 2018, they exceeded durable goods earnings by 11 percent and became the second-highest earning industry in durables behind transportation equipment. This growth in earnings is consistent with increased demand for technology across all areas of the economy.

Printing and paper manufacturing

In contrast to earnings growth in computer and electronic products, some of the weakest earnings growth between 1990 and 2018 occurred in printing and related support activities (1.9 percent annually) and paper and paper products (2.2 percent). In addition, employment in printing and paper has declined steadily since 1990. The job losses and weakness in earnings growth can be attributed to falling circulations and revenues of newspapers and other printed media with the advent and growing dominance of digital media.^[13]

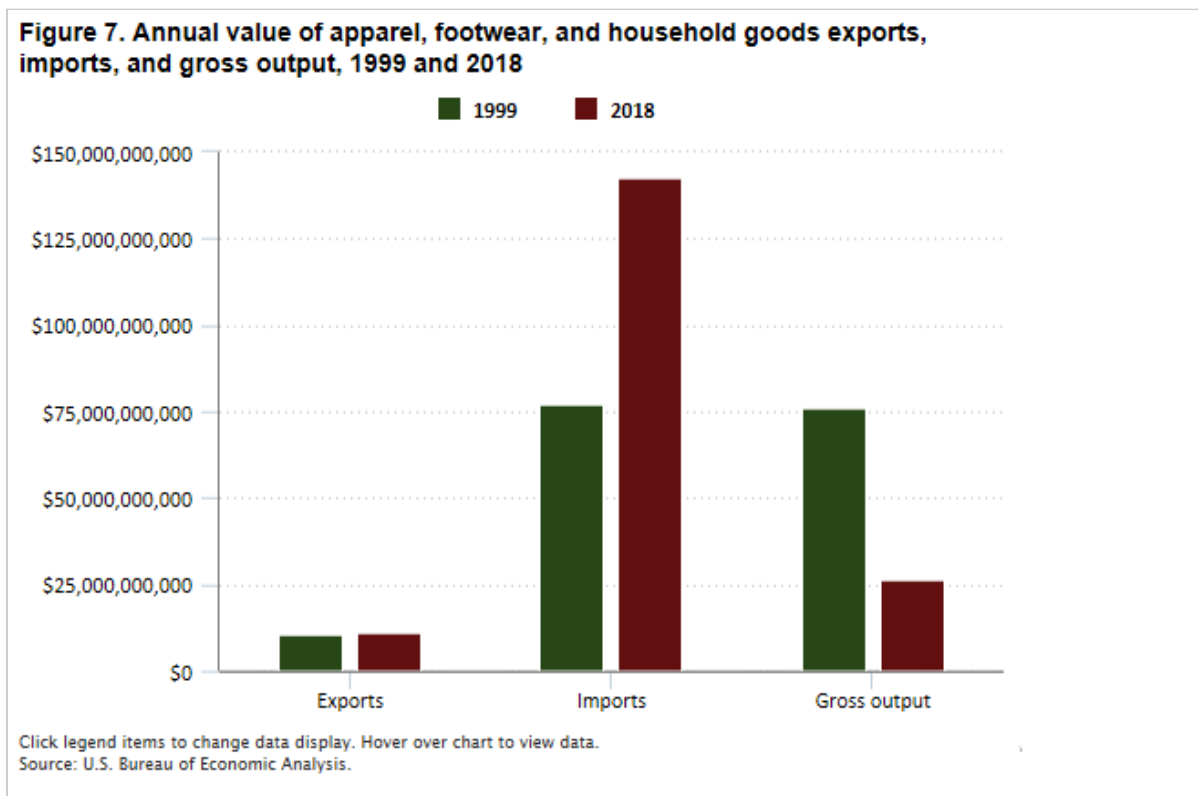
Apparel and textiles

Over the 1990–2018 period, hourly earnings in apparel, textile mills, and textile product mills have ranked the lowest among all of manufacturing's component industries. Both the textile mills and textile product mills industries saw below-average earnings growth; and while apparel manufacturing earnings showed the strongest percentage

growth among all manufacturing industries since January 1990, the level of hourly earnings for apparel workers joined those of textile product mill workers at the bottom of all manufacturing industry earnings.

Employment in each of the apparel and textile industries fell dramatically from January 1990 through the 2007–09 recession. Employment has leveled off in textile product mills since then, and employment losses in textile mills and apparel have decelerated through 2018. Apparel experienced the largest decrease in employment among all manufacturing component industries over the 1990–2018 period, falling by 91 percent. The substantial loss of jobs in apparel manufacturing over the period more than offset gains made in hourly earnings, with aggregate weekly payrolls of apparel workers in 2018 representing just 27 percent of 1990 pay levels.^[14]

Rising competition from apparel imports has driven down both employment and the number of apparel manufacturers in the United States. Over the 2001–2018 period, the number of apparel manufacturers declined from more than 15,000 establishments to fewer than 7,000.^[15] In addition, the trade deficit for apparel, footwear, and household goods doubled over the period, to \$132 billion, and domestic output of apparel fell from \$76 billion in 1999 to \$27 billion in 2018.^[16] (See figure 7.)



Summary

Although manufacturing industries held a reputation for stable, well-paying jobs for much of the 20th century, shifts within the industry in the last several decades have altered that picture considerably. Since 1990, trends in average hourly earnings across manufacturing industries have been disparate. While a few industries have shown strong earnings growth, many others have shown earnings increasing at a much slower rate than the overall private sector, with motor vehicles and parts demonstrating the most notable slowing. In fact, despite a long history

of surpassing private sector earnings, average hourly earnings in durable goods have recently been eclipsed by the overall private sector average.

While employment and earnings in most industries within manufacturing are susceptible to changes in forces such as trade and the U.S. business cycle, others may be more directly affected by movements in other measures, such as the price of oil and the growing use of technology. Therefore, whether more recent improvements in employment and earnings in specific manufacturing industries endure over the long term may depend on a confluence of developments outside the control of manufacturers.

SUGGESTED CITATION

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NOTES

¹ See, for example, Louis Uchitelle, "The wage that meant middle class," *The New York Times*, April 20, 2008, <https://www.nytimes.com/2008/04/20/weekinreview/20uchitelle.html>.

² Throughout this article, "average hourly earnings" and "hourly earnings" are used interchangeably.

³ For more information on benefits, see the National Compensation Survey page on the U.S. Bureau of Labor Statistics (BLS) website at <https://www.bls.gov/ncs/>.

⁴ Unless otherwise indicated, employment, hours, and earnings data are from the BLS Current Employment Statistics (CES) survey. For more information, see the CES page of the BLS website at <https://www.bls.gov/ces>.

⁵ The CES data collection form for manufacturing defines production employees as workers engaging in assembling, fabricating, janitorial activities, maintenance or repair, materials handling, processing, product development, recordkeeping related to production, shipping or receiving, storage or warehousing, and trucking. For more information, see the CES data collection form at https://www.bls.gov/ces/idcf/formc_sp.pdf.

⁶ Production employees in manufacturing and production and nonsupervisory employees in other industries are referred to as "workers" or "employees," and average hourly earnings, average weekly earnings, and average weekly hours discussed throughout this article pertain to production workers in manufacturing and production and nonsupervisory workers in other industries (including total private).

⁷ For more information, see Angela Clinton, "An average mystery in hours and earnings data entails a weighty explanation," *Beyond the Numbers*, vol. 7, no. 9, June 2018, <https://www.bls.gov/opub/btn/volume-7/mystery-in-average-of-hours-and-earnings.htm#ednref1>.

⁸ Throughout this article, "average weekly earnings" and "weekly earnings" are used interchangeably.

⁹ Average weekly hours are based on hours paid during the reference period, which is the pay period that includes the 12th day of the month, including any paid leave.

¹⁰ See Kristin Dziczek, "Bob Fish memorial automotive and economic outlook luncheon," *2011 Detroit 3—UAW labor contracts*, Center for Automotive Research, January 19, 2012, <https://www.chicagofed.org/forms/rss/~media/1ca75a4a11c341fa877776a07e0cb501.ashx>.

¹¹ See Jane Slaughter, “UAW agrees to givebacks at Ford,” *Labor Notes*, February 20, 2009, <https://labornotes.org/2009/02/uaw-agrees-givebacks-ford>.

¹² See Sean McAlinden, Kristin Dziczek, and Art Schwartz, “CAR breakfast briefing,” *2011 Detroit 3—UAW labor contracts*, Center for Automotive Research, November 29, 2011, http://www.cargroup.org/wp-content/uploads/2017/02/2011-Detroit-3_UAW-Labor-Contract-Negotiations.pdf.

¹³ See Michael Barthel, “Despite subscription surges for largest U.S. newspapers, circulation and revenue fall for industry overall,” *Fact Tank: News in the Numbers*, June 1, 2017, <https://www.pewresearch.org/fact-tank/2017/06/01/circulation-and-revenue-fall-for-newspaper-industry/>.

¹⁴ Aggregate weekly payrolls are the product of employment and average weekly earnings.

¹⁵ The number of establishments is from the BLS Quarterly Census of Employment and Wages (QCEW); see the QCEW page on the BLS website at <https://www.bls.gov/cew/>.

¹⁶ For more information on the U.S. trade balance, see “U.S. international trade in goods and services, September 2019,” U.S. Bureau of Economic Analysis, <https://www.bea.gov/data/intl-trade-investment/international-trade-goods-and-services>.

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70 years of the Occupational Outlook Handbook

From its start in 1949 as a gathering of reports mainly intended to assist veterans to today's popular online guide, the Occupational Outlook Handbook has been updated and improved to benefit students, career seekers, and others looking for career guidance.

The Bureau of Labor Statistics (BLS) *Occupational Outlook Handbook* turned 70 years old in 2019.^[1] Also known as the “OOH” or just the “*Handbook*,” this mainstay of career guidance has been an important source for people seeking information about careers since its inception.

The *OOH* has had several changes over its 70-year history, from its beginning as a book sold by the U.S. Government Printing Office (GPO), now known as U.S. Government Publishing Office, to its current incarnation as a web-only publication. It was updated irregularly in the beginning but settled into a 2-year cycle in 1966. When BLS launched its website in January 1995, the *OOH* was posted online shortly thereafter. However, the content was a replica of the printed *Handbook*. By 2008, the idea of a redesigned *Occupational Outlook Handbook* took root—an *OOH* that would be more in tune with the way that people search for, read, and use information in the digital age. The redesigned, online-only *OOH* was published in 2012 on the BLS website. The *OOH* remains the most popular product on the BLS website in terms of page views. From June to August 2019, *OOH* pages had 6.5 million page views, or about one-quarter of all page views of the BLS website.^[2]

This article describes how and why the *OOH* has changed since its first publication in 1949. It details how BLS went about redesigning the *Handbook*, considering user needs and preferences while still maintaining the important information on which its users depend.

The *Occupational Outlook Handbook* in print

The first *OOH* took more than 10 years to develop. The impetus was a 1938 report from the Advisory Committee on Education, a committee established by President Franklin D. Roosevelt. The report recommended that an occupational outlook service be created to provide employment outlooks by occupation and that BLS carry out that



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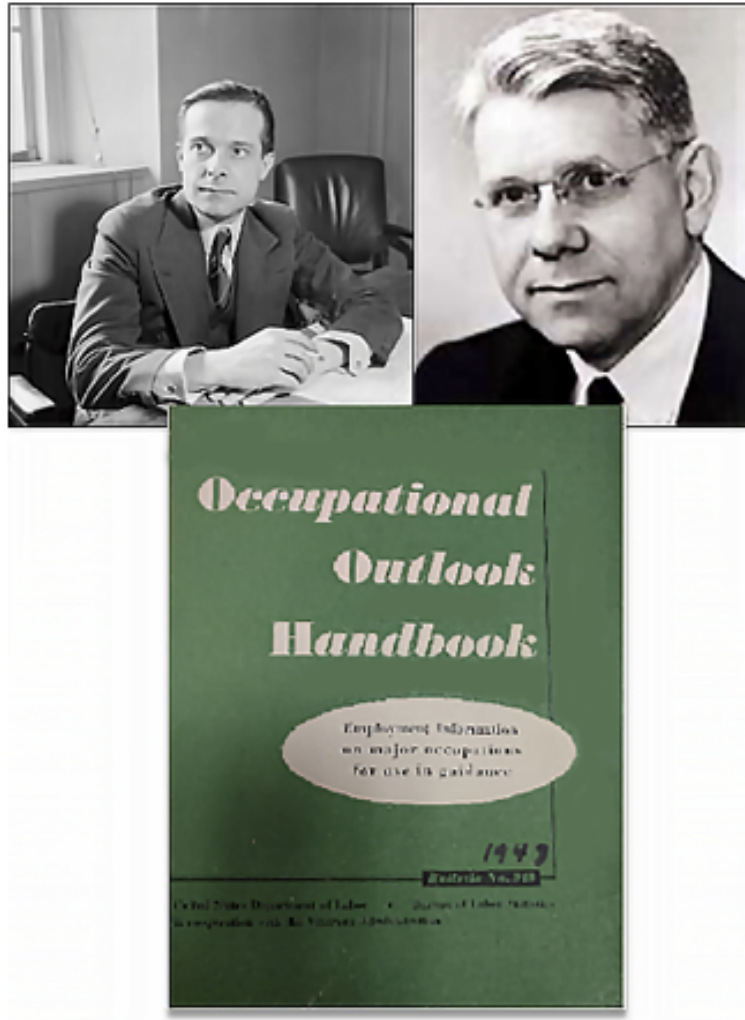
service.^[3] Acting BLS Commissioner A. Ford Hinrichs requested funding from Congress to start the program, and it was provided in 1940. The Veterans Administration (VA), renamed the Department of Veterans Affairs in 1989, also provided funding in the early 1940s to supplement BLS resources. The early research from BLS consisted of short occupational reports, which the VA used to create a manual by punching holes in the reports and binding them together with shoelaces.

BLS began publishing employment outlook bulletins on specific occupations in 1945. In the late 1940s, BLS continued to research occupations and also worked to understand user needs by attending vocational guidance conventions and visiting school counselors.

In 1949, BLS published the first edition of the *OOH*, under BLS Commissioner Ewan Clague.^[4] This first *OOH* covered 288 occupations, which were chosen mainly because the VA thought they would be of interest to veterans.^[5] The GPO sold 40,000 copies of this 1949 *Handbook*, which former BLS Associate Commissioner Harold Goldstein wrote was “astonishing.”^[6]

BLS continued to produce periodic versions of the *Handbook*, typically at 2-year intervals, and created the last print edition (sold by the GPO) in January 2010. (See figure 1 for images of the BLS commissioners and the first *OOH* cover.)

Figure 1. Acting BLS Commissioner A. Ford Hinrichs, January 1940–July 1946; BLS Commissioner Ewan Clague, August 1946–September 1965; and the first *Occupational Outlook Handbook* cover



Sources: A. Ford Hinrichs, Library of Congress, <http://www.loc.gov/pictures/item/2016875522/>, and Ewan Clague, <https://www.bls.gov/bls/history/commissioners/clague.htm>.

Notable changes over the print edition years

From its first publication in 1949 through the last print edition BLS published in January 2010, *OOH* occupations changed, as did the sources of the data used for employment and wages. The *OOH* added several new occupations over the years, while others were eliminated.

Occupational coverage

Until the early 1980s, the occupations covered in the *Handbook* were chosen regardless of whether reliable statistical data existed on them. Criteria for occupational coverage included whether the occupations were large in terms of employment, required a lot of training, or were simply thought to be of interest. Budget availability and staff resources also affected the number of occupations covered in the *OOH*.^[7]

Some occupations dwindled in importance over time (or disappeared from the economy), and new occupations emerged.^[8] Occupations covered in the 1949 edition included many that do not exist today, such as fur blenders, blacksmiths, typewriter servicemen, and hostlers. Occupations such as these were removed from the *Handbook* as demand for their services in the economy lessened and employment declined. The *OOH* gained new occupations as the use of technology grew and computer-related occupations emerged. Computer programmers, for example, first appeared in the *OOH* in 1959 and systems analysts about a decade later.^[9]

Gender stereotyping

The first *OOH* included occupations with gender-specific titles, such as lay-out men and the above-mentioned typewriter servicemen; airplane hostesses and flight stewards; and chainmen, rodmen, and axmen. The occupational reports also included the pronouns “he” or “she” for occupations dominated by men or women and had statements such as “Men serving as ship operators must hold first- or second-class radio telegraph licenses issued by the Federal Communications Commission.” Many reports even discussed work opportunities by gender. For example, the outlook section for chiropractors said, “Opportunities for women appear to be good, as many women prefer to go to members of their own sex for treatment.”^[10]

Beginning with the 1957 *OOH*, BLS set a goal to have more information about women’s employment in the *Handbook*. However, BLS stopped mentioning women’s employment in the *OOH* by the mid-1970s to avoid a perception of stereotyping by gender, since several occupations were dominated by women or else had very few women employed.^[11] BLS mostly adopted non-gender-specific titles with the release of the 1976–77 *OOH*. Any remaining gender-specific titles in the *OOH* were eliminated when BLS moved to the Standard Occupational Classification system, which is used by federal agencies to collect and analyze data by occupation.

Data sources

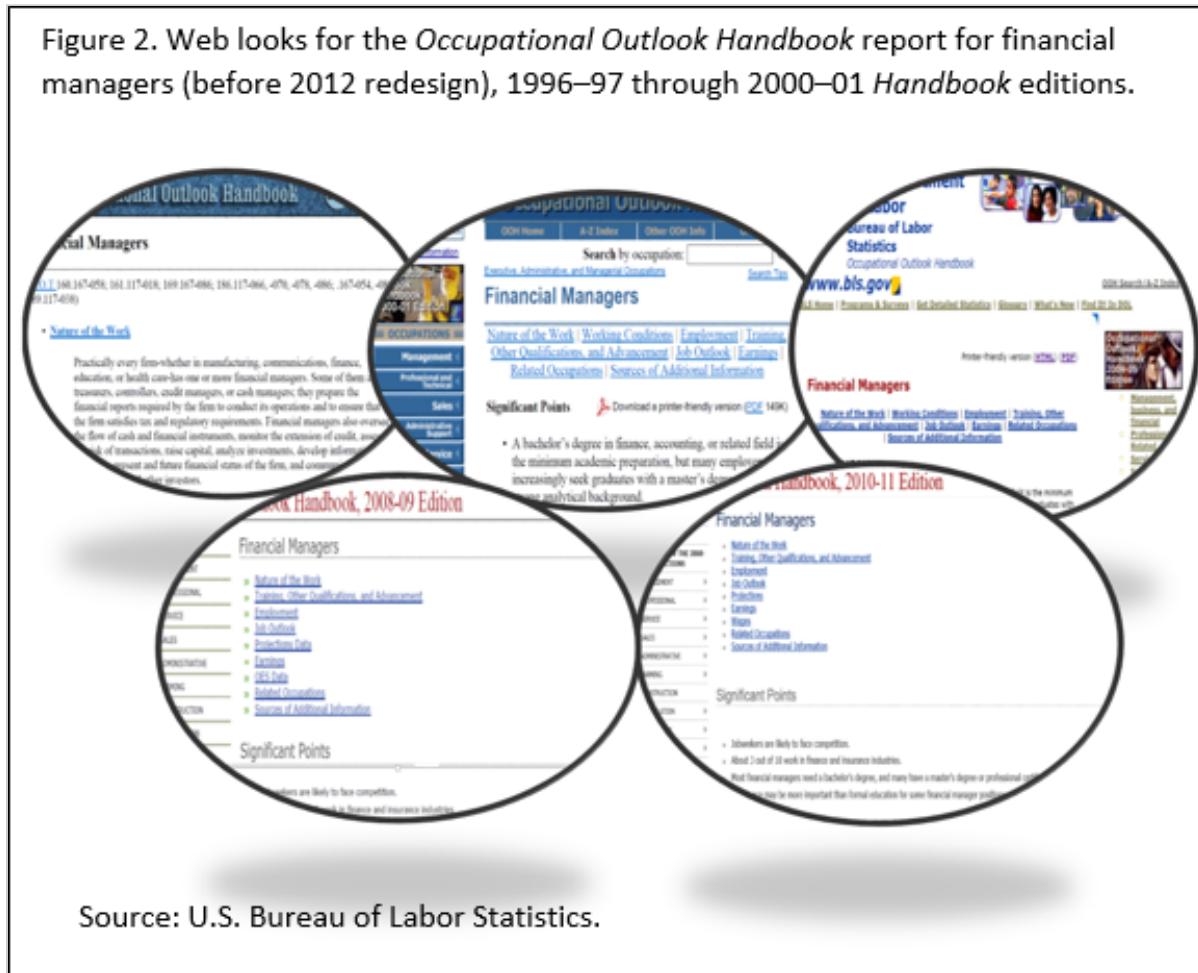
Beginning with the release of the 1984–85 *OOH*, occupations covered were required to have employment data from the Occupational Employment Statistics (OES) survey.^[12] Prior to this time, if no employment data were available, BLS relied on estimates from various associations or organizations that represented occupations. Other government agencies or bureaus produced some material.

The *OOH* used various data on wages over the years, including data from the Current Population Survey, the Current Employment Statistics survey, and other sources. The sources were not consistent because the data were not available for all occupations. With the 2000–01 edition, the *OOH* began using median wage data from the OES survey. The OES data are still used today.

The first web presence

BLS launched its first website in January 1995 at www.bls.gov. The 1996–97 edition of the *OOH* was added to the BLS website about 6 months later. Although the layout of the *OOH* pages on the website changed a few times,

what BLS posted on its website prior to the 2012 redesign was a replica of printed pages in the book, and the book was still being sold by the GPO and other vendors. (See figure 2 for images of OOH looks for financial managers.)



Redesigning the *OOH* for the web

Modernizing the *OOH* was an important part of the 2008 strategic plan for the BLS Employment Projections (EP) program. Identifying and meeting customer needs was a goal of the strategic plan, and the EP program knew that the *OOH* needed to be revamped to better meet those needs. Since the online version at the time was simply a replica of text from the book, BLS was not taking advantage of ways to present the information on the web. Reinventing the *OOH* was therefore a high priority of the BLS, and the redesign was a goal of the overall BLS strategic plan released in November 2010.^[13]

Evaluating stakeholder needs and preferences

BLS collected extensive feedback from external stakeholders to create a plan for the redesign. BLS collected this information via several methods, including through focus groups with counselors; usability tests with students, adult workers, and job seekers; and interviews with technical users. Some of the feedback was collected during sessions at the BLS cognitive laboratory at its headquarters in Washington, DC. Usability tests included observing

participants completing various tasks, recording the time participants needed to complete the tasks, and having the participants rate the difficulty of the tasks.

In addition, BLS reviewed the results of the American Customer Satisfaction Index (ACSI), which was based on a survey conducted by ForeSee. The survey included questions about content, functionality, look and feel, navigation, search, and site performance.

BLS also reviewed *OOH* product use to evaluate demand for *OOH* information and how users were getting that information. BLS analyzed (1) traffic to the *OOH* website, (2) trends in print sales of the *Handbook*, and (3) requests for information received by telephone, regular mail, and email.

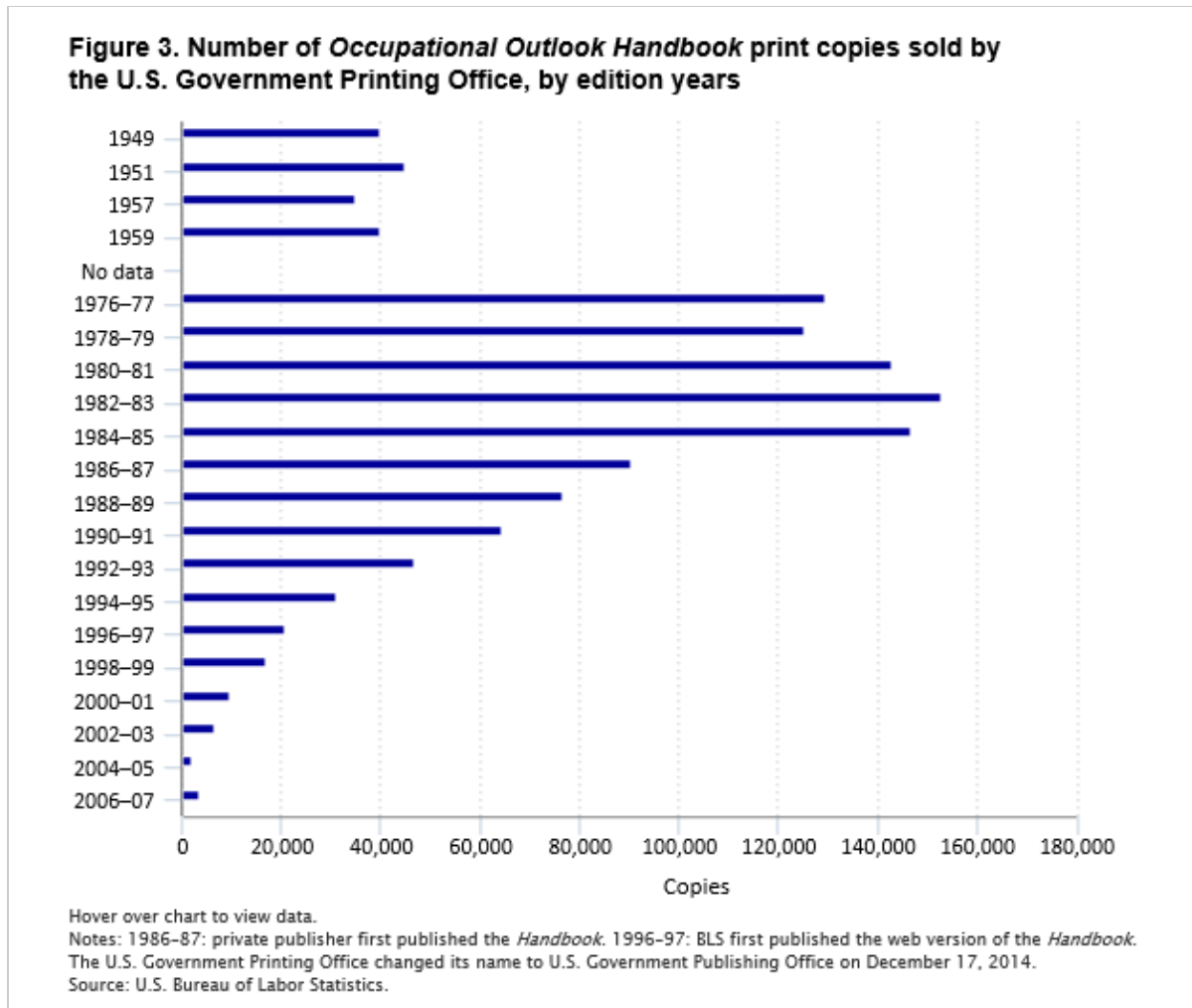
Redesign focus

The information gathered from stakeholders provided many goals for the redesign, including the following:

- An updated and more modern look
- Improved search functions, navigation, and organization
- Easily scanned information that uses bullets and summaries
- Plain language text that is written for the web
- Understandable headings and terminology
- Data for states and local areas
- Web-specific design, such as graphics, photographs, and charts
- Continuously updated information

A team of BLS staff members, consisting of economists, a survey methodologist, a research psychologist, and information technology experts, redesigned the *Handbook*.^[14] The team prepared prototypes of *OOH* pages, including the “landing pages” (pages that serve as gateways to other pages), the home page, and the occupational profile pages themselves. Usability tests of the prototypes were conducted with students, working adults, and counselors, including attendees at national conferences for counselors. In addition, BLS continued to collect feedback on the various prototypes through focus groups and card-sorting exercises.

BLS decided to devote resources to developing the redesigned *OOH* as a web-only publication and to no longer produce pages for a print version to be sold by the GPO. Sales of the GPO-printed *Handbook* had been declining dramatically since about the mid-1980s, which was when the *OOH* was first published by a private publisher. Sales continued to decline as the *OOH* was published on the web about 10 years later. (See figure 3.) BLS surveyed a number of users, including individuals and organizations who had purchased a copy from the GPO, attendees at professional counselor conferences, and other audiences. Analyses indicated that students preferred an online version and that counselors also preferred it. BLS determined that private publishers could continue to meet the demand for a printed book.



BLS decided to have the redesigned *OOH* produced via a content management system (CMS), and it was the first BLS product to be published with this system. A CMS allows staff to review and edit content through a flow process and produces HTML for web pages. It allows the authors and reviewers to immediately preview the webpages before approving them for publication. BLS also produced a mobile version of the *OOH* using the CMS, which was important in modernizing the guide. The mobile version optimized the appearance and usage of the *OOH* on mobile phones and tablets.

Design principles for a web *OOH*

BLS incorporated most of what users wanted and preferred in the redesigned *Handbook* (some enhancements came later; see the section later in this article on improvements since the redesign). The design provides numerous navigation and search options, with better organization of the content. Navigation and search options include occupational groups landing pages and an occupation finder. The occupational groups landing pages display summary information about related occupations, such as entertainment and sports. With the occupation finder, a user can search occupations by combinations of various characteristics: entry-level education, on-the-job training, projections of new jobs and the growth rate, and pay. (See figure 4 for an example of an occupation finder result for occupations that typically need a bachelor's degree for entry.)

Figure 4. Occupation finder search for bachelor's degree occupations projected to add 50,000 jobs or more (from 2018 to 2028) and had median annual wages of \$80,000 or more in 2018

Occupation Finder

Search: Use the drop-down menu in one or more columns to narrow your search.

Sort: Use the arrows at the top of each column to sort alphabetically or numerically.

Showing 1 to 7 of 7 entries (filtered from 808 total entries)

Show 25 entries

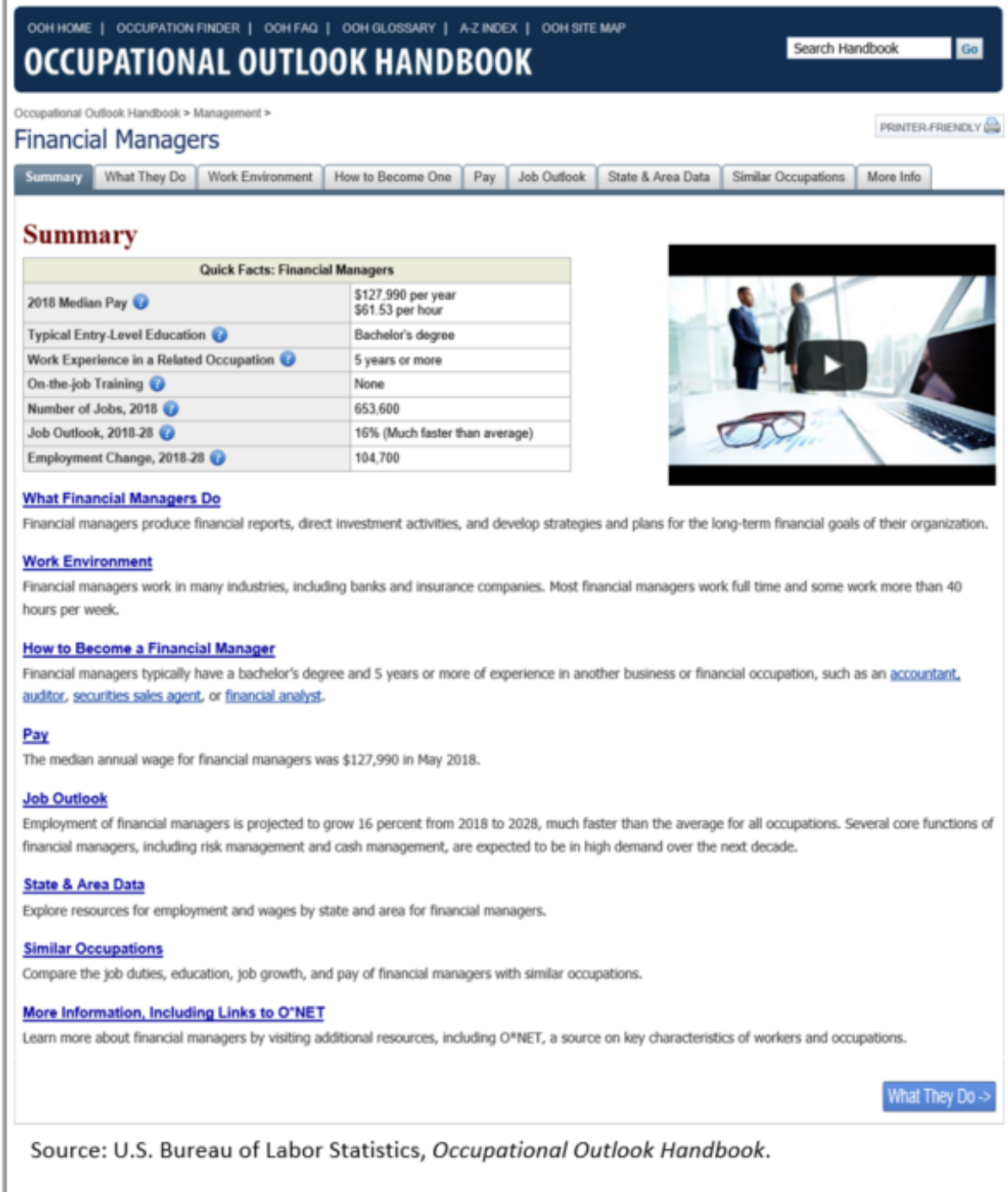
OCCUPATION	ENTRY-LEVEL EDUCATION	ON-THE-JOB TRAINING	PROJECTED NUMBER OF NEW JOBS	PROJECTED GROWTH RATE	2018 MEDIAN PAY
	Bachelor's degree		50,000 or more		\$80,000 or more
Computer systems analysts	Bachelor's degree	None	50,000 or more	Faster than average	\$80,000 or more
Financial managers	Bachelor's degree	None	50,000 or more	Much faster than average	\$80,000 or more
General and operations managers	Bachelor's degree	None	50,000 or more	Faster than average	\$80,000 or more
Management analysts	Bachelor's degree	None	50,000 or more	Much faster than average	\$80,000 or more
Managers, all other	Bachelor's degree	None	50,000 or more	As fast as average	\$80,000 or more
Medical and health services managers	Bachelor's degree	None	50,000 or more	Much faster than average	\$80,000 or more
Software developers, applications	Bachelor's degree	None	50,000 or more	Much faster than average	\$80,000 or more

Source: U.S. Bureau of Labor Statistics, *Occupational Outlook Handbook*.

Additional navigation capabilities include an A–Z index and a content search. The *OOH* home page has links to charts by which users can view the highest paid occupations, fastest growing occupations, and occupations with the most new jobs. The charts provide links to further information (including occupational profiles) for each occupation shown.

The occupational profiles have information displayed in a “tab” format across the page by topic, along with information in bulleted lists, tables, and charts. User testing showed that users preferred a tab format over a long page that required scrolling and also showed that they completed tasks more quickly. Users can easily navigate through a profile by selecting the appropriate tab or by clicking on arrows at the bottom of each page. The occupational profile tabs have understandable headings and are search-engine optimized, which means that they are more likely to appear high on a list of results returned by a search engine. The first tab is a summary page that allows users to quickly scan the occupation. It includes a “Quick Facts” table with key information. Other tabs describe duties, work environment, education and training requirements, pay, and the job outlook for the occupation. The pay and job outlook tabs contain charts that allow users to compare the pay and outlook for an occupation to the total for all occupations and to the major occupation group to which the occupation belongs. The tab for state and area data provides sources of information on state and area employment, projections, and wages (this tab was not part of the initial release of the redesigned *OOH* and was added in 2015). (See figure 5 for an image of the occupational profile for financial managers.)

Figure 5. *Occupational Outlook Handbook* profile for financial managers



Text rewritten for the web

The somewhat dense and lengthy text in the print *OOH* was not suitable for today's readers for a website. Web content needs to be high quality and to retain the important information that users want. To ensure this level of

quality, BLS contracted an outside expert to train and guide staff on writing for the web.^[15] Staff learned a number of key points for successful web writing, including the following:

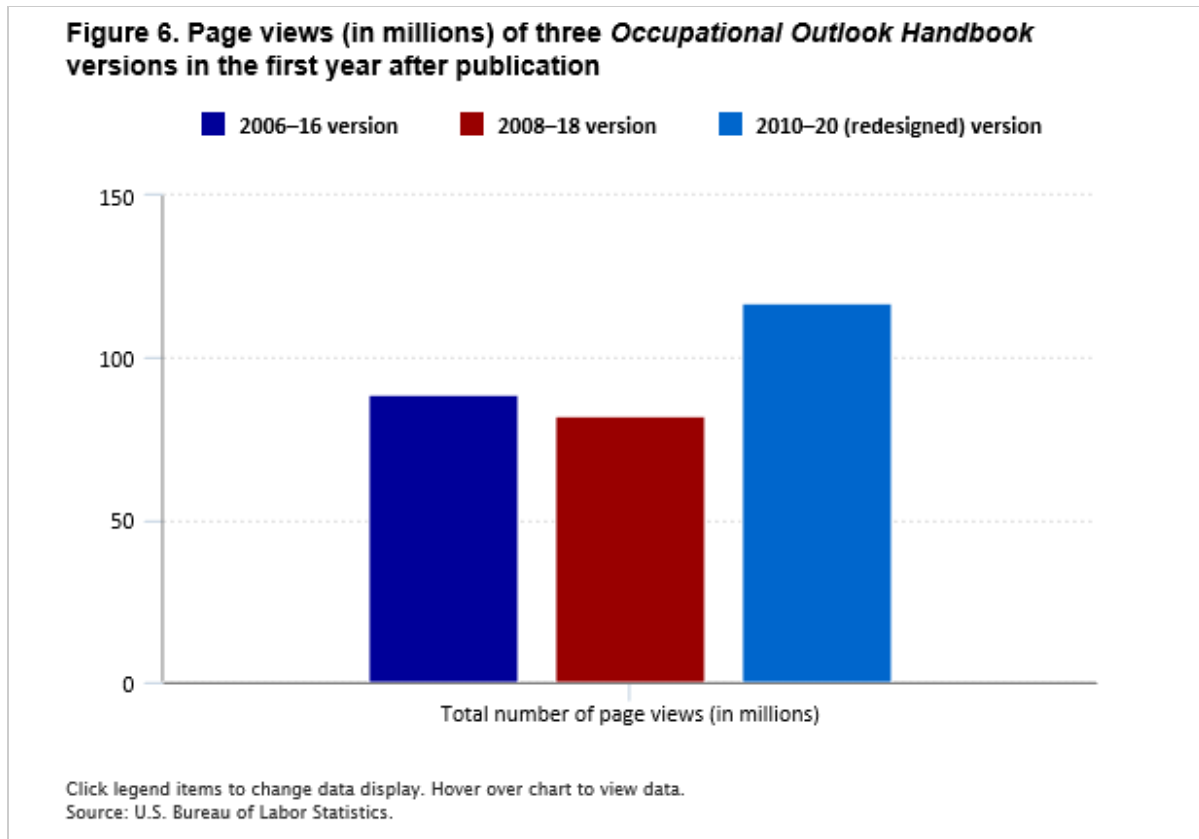
- Ensure the key message or main point comes first.
- Use as few words as possible to get the point across.
- Use the simplest words to express the correct meaning so that all web readers easily understand the text.
- Write short paragraphs and sentences using active voice, if possible.
- Use bullets, which web readers can more easily scan, rather than dense text.

BLS economists prepared 341 occupational profiles (covering 536 occupations), keeping in mind the principles of writing for the web. This effort resulted in more content in the web redesign *OOH* than had been in the print *OOH*. Individual profiles were added for all the science and engineering occupations, which had been covered in groups in the print *OOH*. In addition, new occupational profiles covering various computer and business occupations were added, among others.

The redesigned *OOH* and further improvements

The redesigned *OOH* was released on March 29, 2012, and was based on employment projections for the 2010–20 decade. It included about 50 more occupational profiles than the print version. BLS has subsequently updated the *OOH* with the most recent projections data as they have become available, along with incorporating new research regarding the occupations and updated wage data.

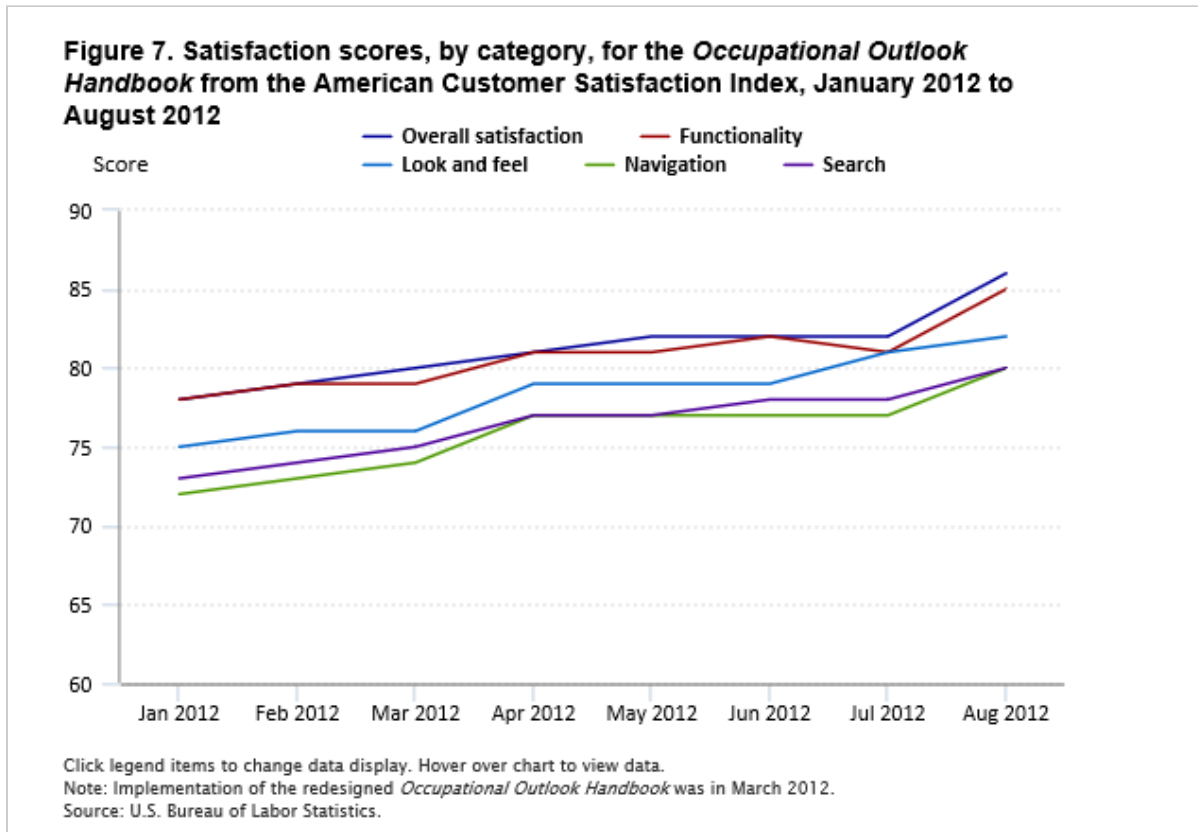
The number of *OOH* page views increased substantially after the redesigned version was released in March 2012. In the 12 months following the release, the number of page views totaled more than 117 million, an increase of about 42 percent over the 12 months after the prior (before the redesign) *OOH* was released. (See figure 6.) Among the reasons for the increased views were the additional occupational profiles and the new mobile site, which itself received 8.9 million views. To help gauge the extent that the redesign had on usage, BLS compared, between the print version and the redesigned version, page views of profiles that had identical occupational coverage. Views of these profiles grew by 10.8 percent.^[16]



BLS also used a survey to evaluate the success of the redesigned *OOH* in meeting customer needs and to determine if anything could be done better. The survey was posted on the BLS website in 2013 and received 2,250 responses. BLS also sought feedback directly from the counselor community. The National Association of College and Employers (NACE) posted a link to the survey on its website, which generated 357 responses.^[17] The NACE member population is different from the general population that completed the survey on the BLS website, since nearly all respondents from NACE indicated they were familiar with the *OOH*, whereas the general population indicated less familiarity.

Respondents to both surveys rated the redesign of the *OOH* highly successful. The response rates from NACE were somewhat higher than those from the general population, which may again reflect the greater familiarity of the NACE members with the *OOH*. The survey asked if the redesigned *OOH* was an improvement over the prior version. Of those users familiar with the *OOH*, 84 percent of the general population felt the redesigned *Handbook* was improved over the previous version (either much improved, somewhat improved, or a little improved), and 91 percent of NACE respondents felt the same. Questions were asked about satisfaction with (1) retaining important information, (2) navigation, (3) attractiveness, and (4) readability. Most of the general population respondents felt the efforts in each area were successful (either very successful or successful), ranging from 68 percent for retaining important information to 75 percent for readability. For the NACE respondents, satisfaction ratings ranged from 80 percent for readability to 87 percent believing the redesign was successful in retaining important information. Of the general population and the NACE respondents, 87 percent and 93 percent, respectively, rated the redesigned *OOH* as an excellent or good source of career information. Through several open-ended questions, both sources suggested improvements.

BLS again analyzed results from the ACSI after the redesigned version was posted. BLS compared results for April to August 2012 with results just prior to the redesign (January to March 2012). During this period, about 4,000 visitors to the *OOH* website completed surveys. Overall satisfaction increased after the redesigned version was posted. The ratings for the drivers of satisfaction—search, functionality, look and feel, and navigation—all grew.^[18] (See figure 7.)



Since redesigning the *OOH*, BLS has improved the website further on the basis of user feedback. As noted earlier, a “State & Area Data” tab was added to all occupational profiles in December 2015 to assist customers find information specific to states and local areas. The *OOH* was updated with the most current OES wage data in spring 2016. Updating the *OOH* with current data as soon as the data were released is a change from the previous practice of making the wage update at the same time as the projections release, which was a legacy from producing the print version. BLS will continue to update the wages in the *Handbook* as soon as the data are released. In addition, BLS added career videos to occupational profiles beginning in 2017.^[19] Video content was produced by the U.S. Department of Labor’s CareerOneStop in consultation with BLS subject matter experts. Nearly all profiles in the *OOH* currently have a career video embedded on their summary pages.

Recent and upcoming enhancements

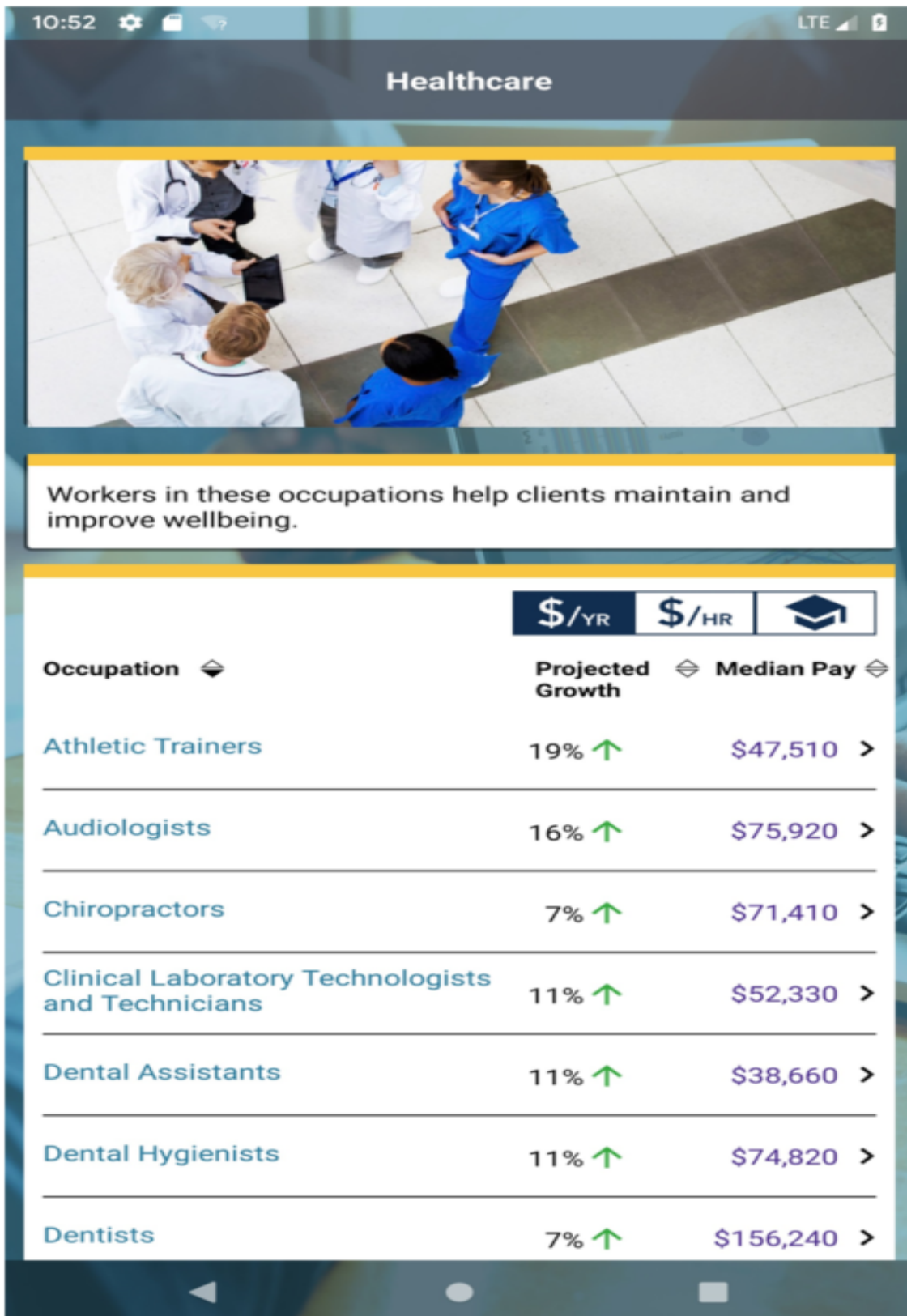
BLS continues to strive to improve its products, including the projections data that form the basis of the *Handbook*. BLS will begin releasing projections data annually to provide more timely information. The last set of biennial employment projections (for the 2018–28 decade) was released in September 2019. BLS plans to release its first

set of annual projections, for the 2019–29 period, in 2020. The *OOH* will continue to incorporate the latest projections as they are released.

In addition, BLS continues to strive to provide timely information by keeping all occupational profiles up to date. Not only are the projections and wage data incorporated as soon as possible but also other updates are being published when ready. For example, when licensure requirements change for an occupation, BLS does not have to wait for a specific publication date to update information, as was the case when the *OOH* was a printed volume. BLS now revises the *OOH* as soon as the research is complete.

BLS released an *OOH* app (application) in fall 2019.^[20] Many people access the internet using a smartphone, which is the main source of internet access for some. According to a Pew Research Center study, 81 percent of Americans have a smartphone, with 96 percent of people ages 18 to 29 owning one.^[21] This age group is a key demographic for the *OOH*. Pew also notes that the percentage of people who use smartphones to access the internet has been growing for all age groups.^[22] The *OOH* app will provide an additional avenue for all users to access career information and will let users personalize the app by filtering searches and by “liking,” saving, viewing, and comparing favorites. (See figure 8 for an image of the app.)

Figure 8. Image from the *Occupational Outlook Handbook* app



Source: U.S. Bureau of Labor Statistics.

BLS is planning an upcoming enhancement to the *Handbook* using field of degree data from the American Community Survey, administered by the U.S. Census Bureau. The new pages for fields of degree will include information about which occupations are most popular for those with a bachelor's degree in various fields and a link to the relevant occupational profiles, along with data on the employment outlook, median wage, and more. These pages will provide statistics for graduates with specific degree types within a field compared with all graduates in that field. For example, the page for the biology field will have information about specific types of biology degrees, such as general, zoology, and microbiology. The field of degree pages will provide more ways for users to search for occupations of interest to them and navigate between them.

Conclusion

The first semblance of career guidance from BLS was an assortment of pamphlets tied together and distributed by the Veterans Administration in 1946. In 1949, BLS published the first *Occupational Outlook Handbook*, and in 2012, redesigned the *Handbook* as a web publication. BLS continues to serve the customer by regularly improving the *OOH*, keeping it useful and current.^[23] The *OOH* remains a popular and trusted source for career guidance 70 years after its inception.

SUGGESTED CITATION

Teresa L. Morisi, "70 years of the Occupational Outlook Handbook," *Monthly Labor Review*, U.S. Bureau of Labor Statistics, December 2019, <https://doi.org/10.21916/mlr.2019.28>.

NOTES

¹ *Occupational outlook handbook* (U.S. Bureau of Labor Statistics, modified September 4, 2019), <https://www.bls.gov/ooh/>.

² This count does not include views of the mobile version of the *Occupational outlook handbook* (OOH), so actual page views are higher.

³ Harold Goldstein, "The early history of the *Occupational outlook handbook*," *Monthly Labor Review*, May 1999, pp. 3–7, <https://www.bls.gov/opub/mlr/1999/05/art1full.pdf>.

⁴ Ibid.

⁵ *Occupational outlook handbook*, Bulletin no. 940 (U.S. Department of Labor, U.S. Bureau of Labor Statistics, 1949), p. 1.

⁶ Goldstein, "The early history of the *Occupational outlook handbook*," p. 6.

⁷ Michael J. Pilot, "*Occupational outlook handbook*: a review of 50 years of change," *Monthly Labor Review*, May 1999, pp. 8–26, <https://www.bls.gov/opub/mlr/1999/05/art2full.pdf>.

⁸ Occupational classifications and titles changed over the years and included various editions of the *Dictionary of occupational titles* and the Standard Occupational Classification system. For more information, see <https://occupationalinfo.org/> and <https://www.bls.gov/soc/>.

⁹ Pilot, "*Occupational outlook handbook*: a review of 50 years of change."

¹⁰ *Occupational outlook handbook*, Bulletin no. 940, pp. 55, 86.

¹¹ Ibid.

¹² Occupational Employment Statistics is a survey of establishments, which provides employment and wage data, and is conducted by BLS. Currently, data are available for more than 800 occupations.

[13](https://www.bls.gov/bls/bls_strategic_plan_201011.pdf) U.S. Bureau of Labor Statistics, Department of Labor, *Strategic Plan*, November 2010, https://www.bls.gov/bls/bls_strategic_plan_201011.pdf

[14](#) Members of the team included current and past BLS employees Kristina J. Bartsch, Jean E. Fox, Karen Ann Krein, Drew S. Liming, William P. Mockovak, Roger J. Moncarz, Daniel G. Murphy, Jerie C. Refugia, Colleen D. Teixeira-Moffat, Michael G. Wolf, and Jo-Ann L. Yu.

[15](#) Staff received training and guidance from Janice (Ginny) Redish, author of *Letting go of the words, writing web content that works*, 2nd ed. (Elsevier, Inc., Morgan Kaufmann, August 14, 2012), <https://redish.net/books/letting-go-of-the-words/>.

[16](#) William P. Mockovak and Kristina J. Bartsch, “Reinventing and evaluating a redesigned *Occupational outlook handbook*” (U.S. Bureau of Labor Statistics, September 2013), <https://www.bls.gov/osmr/research-papers/2013/pdf/st130010.pdf>.

[17](#) NACE is professional association that connects college career services professionals, university relations and recruiting professionals, and business solution providers. See <https://www.naceweb.org/>.

[18](#) Mockovak and Bartsch. “Reinventing and evaluating a redesigned *Occupational outlook handbook*.”

[19](#) To view the latest videos, go to the Department of Labor website CareerOneStop at <https://www.careeronestop.org/Videos/NewCareerVideos/new-career-videos.aspx>.

[20](#) For more information about downloading the app, see <https://www.bls.gov/ooh/about/app.htm>.

[21](#) *Mobile fact sheet*, “More fact sheets: internet/broadband | social media” (Pew Research Center, June 12, 2019), <https://pewresearch-org-preprod.go-vip.co/pewinternet/fact-sheet/mobile/>.

[22](#) Monica Anderson, “Mobile technology and home broadband 2019” (Pew Research Center, June 13, 2019), <https://pewresearch-org-preprod.go-vip.co/pewinternet/2019/06/13/mobile-technology-and-home-broadband-2019/>.

[23](#) Readers may send any comments or suggestions regarding the OOH or the app to oohinfo@bls.gov.

RELATED CONTENT

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[Korean Occupational Outlook Handbook: first edition](#), *Monthly Labor Review*, May 2000.

[Occupational Outlook Handbook: a review of 50 years of change](#), *Monthly Labor Review*, May 1999.

[The early history of the Occupational Outlook Handbook](#), *Monthly Labor Review*, May 1999.

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